

EL ENG 226B

Applications of Stochastic Process Theory

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Lecture 1

Setup: L^p Spaces and Uniform Integrability

1.1 Where the course is going

Before the machinery, one example of what it is for.

Example 1.1 (Controlled diffusion). Let $X(t)$ denote the state of a system at time t , steered by a control $u_t(\cdot)$ and corrupted by noise $n(t)$:

$$dX(t) = \underbrace{u_t(X(t))}_{\text{control}} dt + \underbrace{dn(t)}_{\text{noise}}.$$

When the noise is Brownian this is

$$dX_t = u_t(X_t) dt + dB_t, \quad X_0 = x.$$

Q. What is $\mathcal{L}(X_t)$?

Example 1.2 (Diffusion models). Suppose we want to sample from a distribution μ that we can only evaluate up to a constant. The theory developed in this course tells us how to choose the control $u_t(\cdot)$ so that the solution $(X_t)_{t \geq 0}$ of the equation above satisfies

$$\mathcal{L}(X_T) = \mu.$$

This is exactly the design problem behind modern generative diffusion models.

Both examples presuppose that $dX_t = u_t(X_t) dt + dB_t$ means something, that it has a solution, and that the solution has a computable law. Supplying those three things is the content of the course. We start from the beginning.

1.2 The underlying probability space

Throughout, fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 1.3 (L^p spaces). For $p \in [1, \infty)$,

$$L^p(\Omega, \mathcal{F}, \mathbb{P}) \equiv L^p := \{X \text{ a random variable} : \mathbb{E}(|X|^p) < \infty\}.$$

Fact 1.4. L^p is a complete normed vector space under

$$\|X\|_p := \left(\mathbb{E}(|X|^p) \right)^{1/p}.$$

Moreover L^2 is a Hilbert space, with inner product

$$\langle X, Y \rangle = \mathbb{E}(XY), \quad \forall X, Y \in L^2.$$

Remark 1.5. Strictly speaking $\|\cdot\|_p$ is only a norm on *equivalence classes* of random variables modulo almost sure equality — $\|X\|_p = 0$ forces $X = 0$ a.s., not everywhere. As usual we suppress the distinction.

Fact 1.6 (Hölder's inequality). If $\frac{1}{p} + \frac{1}{q} = 1$, then

$$\mathbb{E}(|XY|) \leq \|X\|_p \|Y\|_q.$$

1.3 Uniform integrability

Definition 1.7 (Uniform integrability). A collection $(X_\alpha)_{\alpha \in A}$ is *uniformly integrable* (u.i.) if

$$\lim_{x \rightarrow \infty} \sup_{\alpha \in A} \mathbb{E}(|X_\alpha| \mathbf{1}_{\{|X_\alpha| > x\}}) = 0.$$

The point of the definition is that it is exactly the extra hypothesis needed to upgrade almost sure convergence to convergence in L^p .

Theorem 1.8. Let $p \in [1, \infty)$ and let $X, (X_n)_{n \geq 1} \subset L^p$ satisfy $X_n \rightarrow X$ a.s. Then the following are equivalent:

1. $X_n \rightarrow X$ in L^p , i.e. $\lim_{n \rightarrow \infty} \mathbb{E}(|X_n - X|^p) = 0$;
2. $(|X_n|^p)_{n \geq 1}$ is uniformly integrable;
3. $\mathbb{E}(|X_n|^p) \rightarrow \mathbb{E}(|X|^p)$.

Moreover, if either

- (a) there exists $Y \in L^p$ with $|X_n| \leq Y$ for all $n \geq 1$; or
- (b) there exists $q > p$ with $\sup_{n \geq 1} \mathbb{E}(|X_n|^q) < \infty$,

then ①–③ hold.

Proof. Homework. □

Remark 1.9. Condition (a) is dominated convergence and (b) is a moment bound one notch above p ; each is a sufficient condition for uniform integrability that is easier to check in practice than the definition. Almost every “can I exchange a limit and an expectation?” question in this course is answered by one of these two.

1.4 Filtrations

To encode causality — the idea that at time t we know the past but not the future — we introduce filtrations.

Definition 1.10 (Filtration). A *filtration* $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ of $(\Omega, \mathcal{F}, \mathbb{P})$ is a collection of σ -algebras such that

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \quad \forall s \leq t.$$

In discrete time the index set is just $\mathbb{N}$.

Example 1.11 (Generated filtration). Given $(X_n)_{n \geq 0}$, set

$$\mathcal{F}_n = \sigma(\{X_0, \dots, X_n\}), \quad n \geq 0,$$

the smallest σ -algebra containing $\{\omega : X_k(\omega) \leq \alpha\}$ for all $\alpha \in \mathbb{R}$ and $k \in \{0, \dots, n\}$.

1.5 Completeness and the usual conditions

Definition 1.12 (Null sets). The $\mathbb{P}$ -null sets are

$$\mathcal{N} = \{F \in \mathcal{F} : \mathbb{P}(F) = 0\}.$$

Definition 1.13 (Complete probability space). $(\Omega, \mathcal{F}, \mathbb{P})$ is *complete* if $\mathcal{F}$ contains every subset of every $N \in \mathcal{N}$.

Fact 1.14 (Completion). Every probability space can be uniquely completed to $(\Omega, \tilde{\mathcal{F}}, \tilde{\mathbb{P}})$, where

$$\tilde{\mathcal{F}} = \sigma(\{F \triangle E : F \in \mathcal{F}, E \subseteq N \text{ for some } N \in \mathcal{N}\}), \quad \tilde{\mathbb{P}}(F \triangle E) = \mathbb{P}(F).$$

Notation. From now on we always assume $(\Omega, \mathcal{F}, \mathbb{P})$ is complete.

Definition 1.15 (Standard filtration). A continuous-time filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ is:

- (a) *complete* if $\mathcal{F}_0$ contains all $\mathbb{P}$ -null sets;
- (b) *right continuous* if $\mathcal{F}_t = \bigcap_{s > t} \mathcal{F}_s =: \mathcal{F}_{t+}$ for all $t \geq 0$;
- (c) a *standard filtration* (equivalently, it *satisfies the usual conditions*) if both (a) and (b) hold.

Neither condition is innocuous bookkeeping: completeness is what makes suprema over uncountably many times measurable, and right continuity is what makes hitting times stopping times. We take these up next lecture.

Lecture 2

Filtrations and Conditional Expectation

2.1 Recap

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, completed so that $\mathcal{F}$ contains every $\mathbb{P}$ -negligible set (every subset of a $\mathbb{P}$ -null event). The passage of time is encoded by a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$, where

$\mathcal{F}_t :=$ all information available up to time t = all events we can “measure” up to time t .

In continuous time, $\mathbb{F}$ is a *standard filtration* (equivalently, it satisfies the *usual conditions*) if it is

1. *complete*: $\mathcal{F}_0$ contains all $\mathbb{P}$ -negligible sets (harmless);
2. *right continuous*: $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$ for all $t \geq 0$.

Example 2.1 (Generated filtration). Let $X = (X_t)_{t \geq 0}$ be a random process and

$$\mathcal{F}_t = \sigma(X_s; 0 \leq s \leq t),$$

the smallest σ -algebra with respect to which all of the past is measurable. Then $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ is the *filtration generated by X* .

Fact 2.2. *The completion of the natural Brownian filtration is a standard one.*

2.2 Motivation for the usual conditions

① Completeness makes suprema measurable

Let $(X_t)_{t \geq 0}$ be a process with a.s. continuous sample paths and set

$$S_t = \sup_{0 \leq s \leq t} X_s.$$

A priori this is *not* a random variable: the supremum is over uncountably many indices, so there is no reason for $\{S_t \leq \alpha\}$ to be an event.

Let $\Omega_0 \in \mathcal{F}$ with $\mathbb{P}(\Omega_0) = 1$ be such that $(X_t)_{t \geq 0}$ is continuous on Ω_0 . Then

$$\begin{aligned} \{S_t \leq \alpha\} &= \underbrace{\Omega_0 \cap \{S_t \leq \alpha\}}_{= \Omega_0 \cap \bigcap_{q \in \mathbb{Q} \cap [0, t]} \{X_q \leq \alpha\} \in \mathcal{F}} \cup \underbrace{\Omega_0^c \cap \{S_t \leq \alpha\}}_{\subseteq \Omega_0^c = \mathbb{P}\text{-negligible}}. \end{aligned}$$

On Ω_0 the path is continuous, so its supremum over $[0, t]$ agrees with its supremum over the rationals in $[0, t]$ — a countable intersection, hence an event. The leftover piece sits inside a null set, and is therefore an event precisely *because* we completed the space.

② Right continuity makes hitting times stopping times

Definition 2.3 (Adapted). A process $(X_t)_{t \geq 0}$ is *adapted* to $\mathbb{F}$ if X_t is $\mathcal{F}_t$ -measurable for every $t \geq 0$.

Example 2.4. Every process is adapted to the filtration it generates.

Definition 2.5 (Stopping time). A random variable $\tau: \Omega \rightarrow [0, \infty)$ is a *stopping time* with respect to $\mathbb{F}$ if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$.

Example 2.6 (Hitting times). Let $(X_t)_{t \geq 0}$ be adapted to $\mathbb{F}$ with right continuous paths $\mathbb{P}$ -a.s., and define the hitting time of a set B ,

$$\tau_B := \inf\{t \geq 0 : X_t \in B\}.$$

For B open, right continuity gives

$$\{\tau_B < t\} = \bigcup_{q \in \mathbb{Q} \cap [0, t)} \{X_q \in B\} \in \mathcal{F}_t,$$

since a right continuous path entering an open set must already be in it at some earlier rational time. This only yields the *strict* inequality; to convert it into the definition of a stopping time we intersect,

$$\{\tau_B \leq t\} = \bigcap_{n \geq 1} \{\tau_B < t + \frac{1}{n}\} \in \bigcap_{n \geq 1} \mathcal{F}_{t + \frac{1}{n}} = \mathcal{F}_{t+} = \mathcal{F}_t,$$

where the last equality is right continuity of $\mathbb{F}$.

So the usual conditions are what let us treat the two objects we care most about — running suprema and hitting times — as honest random variables.

2.3 Conditional expectation revisited

Let $X \in L^1$ and let Y be a random variable. The conditional expectation of X given Y is the unique random variable such that

1. it is a measurable function of Y ;
2. $\mathbb{E}(\mathbb{E}(X | Y) g(Y)) = \mathbb{E}(X g(Y))$ for every bounded measurable g .

Remark 2.7. Unwinding the two conditions:

$$\begin{aligned} \textcircled{1} & \iff \mathbb{E}(X | Y) \text{ is } \sigma(Y)\text{-measurable;} \\ \textcircled{2} & \iff \mathbb{E}(\mathbb{E}(X | Y) \mathbf{1}_{\{Y \in B\}}) = \mathbb{E}(X \mathbf{1}_{\{Y \in B\}}) \quad \forall B \in \mathcal{B} \\ & \iff \mathbb{E}(\mathbb{E}(X | Y) \mathbf{1}_A) = \mathbb{E}(X \mathbf{1}_A) \quad \forall A \in \sigma(Y). \end{aligned}$$

Both conditions therefore refer to Y only through $\sigma(Y)$, which is what makes the following generalization the right one.

Definition 2.8 (Conditional expectation given a σ -algebra). For a sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, a random variable Z is a *conditional expectation of $X \in L^1$ with respect to $\mathcal{G}$* if $\sigma(Z) \subseteq \mathcal{G}$ and

$$\mathbb{E}(Z \mathbf{1}_G) = \mathbb{E}(X \mathbf{1}_G) \quad \forall G \in \mathcal{G}.$$

Theorem 2.9. For $X \in L^1$ and $\mathcal{G} \subseteq \mathcal{F}$, the conditional expectation of X with respect to $\mathcal{G}$ always exists and is unique modulo almost sure equivalence. It is denoted $\mathbb{E}(X | \mathcal{G})$.

Example 2.10. $\mathbb{E}(X | Y) = \mathbb{E}(X | \sigma(Y))$.

Example 2.11. Let $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ be generated by $(Y_t)_{t \geq 0}$. Then $\mathbb{E}(X | \mathcal{F}_t) = \mathbb{E}(X | (Y_s)_{s \in [0, t]})$.

2.4 Properties

Theorem 2.12. Let $\mathcal{H} \subseteq \mathcal{G}$ be sub- σ -algebras of $\mathcal{F}$ and $X, Y \in L^1$. Then, almost surely:

1. (Tower property) $\mathbb{E}(\mathbb{E}(X | \mathcal{G}) | \mathcal{H}) = \mathbb{E}(X | \mathcal{H})$;
2. (Taking out what is known) if Y is $\mathcal{G}$ -measurable and $XY \in L^1$, then $\mathbb{E}(XY | \mathcal{G}) = Y \mathbb{E}(X | \mathcal{G})$;
3. (Linearity) $\mathbb{E}(aX + bY | \mathcal{G}) = a \mathbb{E}(X | \mathcal{G}) + b \mathbb{E}(Y | \mathcal{G})$;
4. (Monotonicity) $X \leq Y \implies \mathbb{E}(X | \mathcal{G}) \leq \mathbb{E}(Y | \mathcal{G})$;
5. (Hölder) if $\frac{1}{p} + \frac{1}{q} = 1$, then $\mathbb{E}(|XY| | \mathcal{G}) \leq \mathbb{E}(|X|^p | \mathcal{G})^{1/p} \mathbb{E}(|Y|^q | \mathcal{G})^{1/q}$;
6. (Jensen) if φ is convex and $\varphi(X) \in L^1$, then $\varphi(\mathbb{E}(X | \mathcal{G})) \leq \mathbb{E}(\varphi(X) | \mathcal{G})$.

Example 2.13. The tower property with $\mathcal{H} = \sigma(Z)$ and $\mathcal{G} = \sigma(Y, Z)$ reads

$$\mathbb{E}(\mathbb{E}(X | Y, Z) | Z) = \mathbb{E}(X | Z).$$

Lecture 3

Discrete-Time Martingales

3.1 Conditional expectations are uniformly integrable

Lemma 3.1. *If $X \in L^1$, then*

$$\{\mathbb{E}(X \mid \mathcal{G}) : \mathcal{G} \subseteq \mathcal{F} \text{ a sub-}\sigma\text{-algebra}\}$$

is uniformly integrable.

Proof. Fix $x > 0$. By conditional Jensen, $|\mathbb{E}(X \mid \mathcal{G})| \leq \mathbb{E}(|X| \mid \mathcal{G})$, so in particular $\{|\mathbb{E}(X \mid \mathcal{G})| > x\} \subseteq \{\mathbb{E}(|X| \mid \mathcal{G}) > x\}$ and

$$\mathbb{E}\left(|\mathbb{E}(X \mid \mathcal{G})| \mathbf{1}_{\{|\mathbb{E}(X \mid \mathcal{G})| > x\}}\right) \leq \mathbb{E}\left(\mathbb{E}(|X| \mid \mathcal{G}) \mathbf{1}_{\{\mathbb{E}(|X| \mid \mathcal{G}) > x\}}\right) = \mathbb{E}\left(|X| \underbrace{\mathbf{1}_{\{\mathbb{E}(|X| \mid \mathcal{G}) > x\}}}_{=: F_{\mathcal{G}}}\right),$$

the last step because $F_{\mathcal{G}} \in \mathcal{G}$. By Markov's inequality and the tower property,

$$\mathbb{P}(F_{\mathcal{G}}) \leq \frac{\mathbb{E}(\mathbb{E}(|X| \mid \mathcal{G}))}{x} = \frac{\mathbb{E}(|X|)}{x} = O\left(\frac{1}{x}\right) \quad \text{independently of } \mathcal{G}.$$

Finally, $|X|$ is a single integrable random variable, so for every $\varepsilon > 0$ there is $\delta > 0$ with $\mathbb{P}(A) < \delta \implies \mathbb{E}(|X| \mathbf{1}_A) < \varepsilon$. Choosing $x > \mathbb{E}(|X|)/\delta$ makes $\mathbb{P}(F_{\mathcal{G}}) < \delta$ for *every* $\mathcal{G}$, whence $\sup_{\mathcal{G}} \mathbb{E}(|X| \mathbf{1}_{F_{\mathcal{G}}}) \leq \varepsilon$. So the right-hand side tends to 0 uniformly in $\mathcal{G}$ as $x \rightarrow \infty$. $\square$

3.2 Discrete-time martingales

Definition 3.2 (L^p -martingale). For $p \geq 1$, a process $(X_n)_{n \geq 0}$ is an L^p -martingale with respect to $\mathbb{F}$ if

1. $X_n \in L^p$ for all $n \geq 0$;
2. $(X_n)_{n \geq 0}$ is adapted to $\mathbb{F}$;
3. $\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = X_n$ for all $n \geq 0$.

It is a *submartingale* if “=” in ③ is replaced by “ $\geq$ ”, and it is L^p -bounded if $\sup_{n \geq 0} \mathbb{E}(|X_n|^p) < \infty$.

Example 3.3 (Sums of independent centered variables). Let $(Z_n)_{n \geq 1} \subset L^p$ be independent with $\mathbb{E}(Z_i) = 0$, and set $X_0 = 0$, $X_n = \sum_{i=1}^n Z_i$. Then $(X_n)_{n \geq 0}$ is an L^p -martingale with respect to the filtration generated by Z .

Example 3.4 (Doob martingale). Let $X \in L^p$ and set $X_n := \mathbb{E}(X \mid \mathcal{F}_n)$, $n \geq 0$. Then $(X_n)_{n \geq 0}$ is an L^p -bounded martingale — and, by Lemma 3.1, uniformly integrable.

Lemma 3.5. *Let φ be convex, $(X_n)_{n \geq 0}$ a martingale, and $\varphi(X_n) \in L^p$ for all $n \geq 0$. Then $(\varphi(X_n))_{n \geq 0}$ is an L^p -submartingale. If instead $(X_n)_{n \geq 0}$ is a submartingale and φ is convex and nondecreasing, the same conclusion holds.*

Proof. Conditional Jensen: $\mathbb{E}(\varphi(X_{n+1}) \mid \mathcal{F}_n) \geq \varphi(\mathbb{E}(X_{n+1} \mid \mathcal{F}_n)) = \varphi(X_n)$. In the submartingale case, $\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) \geq X_n$ and monotonicity of φ gives $\varphi(\mathbb{E}(X_{n+1} \mid \mathcal{F}_n)) \geq \varphi(X_n)$. $\square$

Example 3.6. If $(X_n)_{n \geq 0}$ is an L^p -martingale then $(|X_n|^p)_{n \geq 0}$ is an L^1 -submartingale.

3.3 Martingale transforms

Definition 3.7 (Previsible). $(C_n)_{n \geq 1}$ is *previsible* if C_n is $\mathcal{F}_{n-1}$ -measurable for every $n \geq 1$.

Definition 3.8 (Martingale transform). For $C = (C_n)_{n \geq 1}$ and $X = (X_n)_{n \geq 0}$, define

$$(C \cdot X)_n = \sum_{i=1}^n C_i (X_i - X_{i-1}), \quad (C \cdot X)_0 = 0.$$

Lemma 3.9. *Let C be previsible and bounded. Then:*

- (a) *if X is an L^p -martingale, so is $C \cdot X$;*
- (b) *if X is an L^p -submartingale and $C \geq 0$, then $C \cdot X$ is an L^p -submartingale.*

Remark 3.10. This is the statement to read twice: once with every parenthetical dropped, once with all of them kept. The nonnegativity of C is exactly what is needed in the submartingale case and exactly what is *not* needed in the martingale case — a negative bet reverses an inequality but preserves an equality.

The gambling reading is worth keeping in mind: C_i is the stake placed on round i , chosen knowing only $\mathcal{F}_{i-1}$, and $(C \cdot X)_n$ is the accumulated fortune. No previsible betting strategy turns a fair game into a favorable one.

3.4 Stopped processes

Notation. $\tau: \Omega \rightarrow \mathbb{N} \cup \{+\infty\}$ is a stopping time if $\{\tau \leq n\} \in \mathcal{F}_n$ for all $n \geq 0$, and $a \wedge b := \min(a, b)$.

Given a process $(X_n)_{n \geq 0}$ and a stopping time τ , the *stopped process* is $(X_{n \wedge \tau})_{n \geq 0}$, i.e.

$$X_{n \wedge \tau} = \begin{cases} X_n & \text{if } n \leq \tau, \\ X_\tau & \text{if } n > \tau. \end{cases}$$

Theorem 3.11. *If $(X_n)_{n \geq 0}$ is an L^p -(sub)martingale and τ is a stopping time, then $(X_{n \wedge \tau})_{n \geq 0}$ is also an L^p -(sub)martingale.*

Proof. Choose

$$C_n = \mathbf{1}_{\{\tau \geq n\}} = 1 - \mathbf{1}_{\{\tau \leq n-1\}} \in \mathcal{F}_{n-1},$$

which is previsible, bounded, and nonnegative. Then

$$(C \cdot X)_n = \sum_{i=1}^n \mathbf{1}_{\{\tau \geq i\}} (X_i - X_{i-1}) = X_{n \wedge \tau} - X_0,$$

so the claim follows from Lemma 3.9. $\square$

Theorem 3.12. *Let $(X_n)_{n \geq 0}$ be an L^1 -submartingale and τ a stopping time. Then*

$$\mathbb{E}(X_0) \leq \mathbb{E}(X_{n \wedge \tau}) \leq \mathbb{E}(X_n) \quad \forall n \geq 0.$$

Proof. By Theorem 3.11, $(X_{n \wedge \tau})_{n \geq 0}$ is a submartingale, so

$$\mathbb{E}(X_{n \wedge \tau}) \geq \mathbb{E}(X_{0 \wedge \tau}) = \mathbb{E}(X_0),$$

which is the left-hand inequality. For the right-hand one, set

$$X'_n := X_n - X_{n \wedge \tau}.$$

Claim 3.13. *$(X'_n)_{n \geq 0}$ is a submartingale.*

It is adapted and L^1 , both inherited from X and X^τ . For the submartingale inequality, note

$$X'_{n+1} = X'_n + \mathbf{1}_{\{\tau \leq n\}} (X_{n+1} - X_n),$$

since the increment of X and the increment of X^τ agree on $\{\tau > n\}$ and the latter vanishes on $\{\tau \leq n\}$. Hence

$$\begin{aligned} \mathbb{E}(X'_{n+1} \mid \mathcal{F}_n) &= \mathbb{E}\left(X'_n + \mathbf{1}_{\{\tau \leq n\}} (X_{n+1} - X_n) \mid \mathcal{F}_n\right) \\ &= X'_n + \underbrace{\mathbf{1}_{\{\tau \leq n\}}}_{\geq 0} \underbrace{(\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) - X_n)}_{\geq 0} \geq X'_n, \end{aligned}$$

using that $\mathbf{1}_{\{\tau \leq n\}}$ is $\mathcal{F}_n$ -measurable. This proves the claim.

Consequently $\mathbb{E}(X'_n) \geq \mathbb{E}(X'_0) = 0$, i.e. $\mathbb{E}(X_n) \geq \mathbb{E}(X_{n \wedge \tau})$. $\square$

Remark 3.14. Taking X to be a martingale gives both inequalities with equality: $\mathbb{E}(X_{n \wedge \tau}) = \mathbb{E}(X_0)$ for every n . The content of the optional stopping theorem is knowing when one may let $n \rightarrow \infty$ here — and that is precisely a uniform integrability question, which is why Lemma 3.1 came first.

Lecture 4

Optional Stopping and Doob's Inequalities

4.1 Optional stopping

Theorem 4.1 (Optional stopping, bounded form). *Let $(X_n)_{n \geq 0}$ be a submartingale and let τ be a stopping time. Then*

$$\mathbb{E}(X_0) \leq \mathbb{E}(X_{n \wedge \tau}) \leq \mathbb{E}(X_n) \quad \forall n \geq 0. \quad (\star)$$

Remark 4.2. Equality throughout if $(X_n)_{n \geq 0}$ is a martingale.

Q. Stopped submartingales are still submartingales, so $n \mapsto \mathbb{E}(X_{n \wedge \tau})$ is nondecreasing. Why isn't $(\star)$ obviously true?

The first inequality *is* immediate from that observation: it just says $\mathbb{E}(X_{0 \wedge \tau}) \leq \mathbb{E}(X_{n \wedge \tau})$. The second is the content of the theorem, and it compares two *different* processes — $(X_n)_{n \geq 0}$ and the stopped process $(X_{n \wedge \tau})_{n \geq 0}$ — at the same time n .

The result is Doob's stopping theorem. One might hope to infer $\mathbb{E}(X_\tau) = \mathbb{E}(X_0)$ from it, but that is *not* true in general.

Example 4.3 (Random walk stopped at -1). Let $X_n = \sum_{i=1}^n Z_i$ with Z_i i.i.d. $\text{Unif}\{-1, 1\}$, so $X_0 = 0$, and let

$$\tau := \inf\{n \geq 0 : X_n = -1\}.$$

Then $\tau < \infty$ a.s., and

$$\mathbb{E}(X_0) = 0, \quad \mathbb{E}(X_\tau) = -1.$$

So passing to the limit in $(\star)$ is not automatic. Note also that $(X_n)_{n \geq 0}$ and $(X_{n \wedge \tau})_{n \geq 0}$ are different processes.

4.2 Gambler's ruin

Example 4.4 (Exit probabilities). When $(X_n)_{n \geq 0}$ first exits $(-A, B)$, with what probability does it hit B ?

Let $\tau := \inf\{n \geq 0 : X_n \in \{B, -A\}\}$, so that

$$X_\tau = \begin{cases} B & \text{w.p. } P_B, \\ -A & \text{w.p. } 1 - P_B. \end{cases}$$

Claim. $\mathbb{E}(X_{n \wedge \tau}) \rightarrow \mathbb{E}(X_\tau)$ as $n \rightarrow \infty$, since $\tau < \infty$ a.s. Indeed $|X_{n \wedge \tau}| \leq A + B$ for all n , so dominated convergence applies. Combined with Theorem 4.1,

$$\mathbb{E}(X_0) = \mathbb{E}(X_\tau) = P_B \cdot B - (1 - P_B)A = 0 \implies P_B = \frac{A}{B + A}.$$

Example 4.5 (Expected exit time). **Q.** What is $\mathbb{E}(\tau)$?

Pick $M_n = X_n^2 - n$; this is a martingale. Optional stopping gives

$$\mathbb{E}(M_{n \wedge \tau}) = \mathbb{E}(M_0) = 0.$$

To pass to the limit: $M_{n \wedge \tau} \rightarrow M_\tau$ a.s. because $\tau < \infty$ a.s., and

$$|M_{n \wedge \tau}| \leq |X_{n \wedge \tau}|^2 + (n \wedge \tau) \leq \max\{A^2, B^2\} + \tau,$$

where τ is dominated by a geometric random variable with success probability $2^{-(A+B)}$ (in any block of $A + B$ consecutive steps the walk exits with at least that probability), hence integrable. So dominated convergence gives $\mathbb{E}(M_\tau) = 0$, i.e.

$$\begin{aligned} \mathbb{E}(M_\tau) &= \underbrace{\mathbb{E}(X_\tau^2)}_{= P_B B^2 + (1 - P_B)A^2} - \mathbb{E}(\tau) = 0, \end{aligned}$$

and therefore

$$\mathbb{E}(\tau) = \frac{A}{B + A} B^2 + \frac{B}{B + A} A^2 = \frac{AB^2 + BA^2}{B + A} = AB.$$

4.3 Doob's inequalities

Definition 4.6 (Maximal process).

$$X_n^* := \max_{0 \leq k \leq n} |X_k|$$

is the *maximal process* associated with $(X_n)_{n \geq 0}$.

Theorem 4.7 (Doob's maximal inequality). *Let $(X_n)_{n \geq 0}$ be a nonnegative submartingale. Then for all $n \geq 0$ and $\lambda > 0$,*

$$\lambda \mathbb{P}\{X_n^* \geq \lambda\} \leq \mathbb{E}(X_n \mathbf{1}_{\{X_n^* \geq \lambda\}}) \leq \mathbb{E}(X_n).$$

Proof. Let $\tau = \inf\{n \geq 0 : X_n \geq \lambda\}$. Since $X \geq 0$ we have $X_n^* = \max_{k \leq n} X_k$, so

$$\{X_n^* \geq \lambda\} = \{\tau \leq n\} = \left\{ \sum_{k=0}^n X_k \mathbf{1}_{\{\tau=k\}} \geq \lambda \right\},$$

using that $X_\tau \geq \lambda$ on $\{\tau = k\}$ and that the sum is 0 on $\{\tau > n\}$. Hence

$$\begin{aligned} \lambda \mathbb{P}\{X_n^* \geq \lambda\} &= \lambda \mathbb{P}\left\{\sum_{k=0}^n X_k \mathbf{1}_{\{\tau=k\}} \geq \lambda\right\} \leq \mathbb{E}\left(\sum_{k=0}^n X_k \mathbf{1}_{\{\tau=k\}}\right) \\ &\leq \mathbb{E}\left(\sum_{k=0}^n \underbrace{\mathbb{E}(X_n | \mathcal{F}_k) \mathbf{1}_{\{\tau=k\}}}_{\in \mathcal{F}_k}\right) = \sum_{k=0}^n \mathbb{E}\left(\mathbb{E}(X_n \mathbf{1}_{\{\tau=k\}} | \mathcal{F}_k)\right) \\ &= \mathbb{E}\left(X_n \underbrace{\mathbf{1}_{\{\tau \leq n\}}}_{=\mathbf{1}_{\{X_n^* \geq \lambda\}}}\right) \leq \mathbb{E}(X_n). \end{aligned}$$

The second inequality is the submartingale property $X_k \leq \mathbb{E}(X_n | \mathcal{F}_k)$ for $k \leq n$. $\square$

Theorem 4.8 (Doob's L^p inequality). *Let $(X_n)_{n \geq 0}$ be a nonnegative submartingale. Then for every $p > 1$,*

$$\|X_n\|_p \leq \|X_n^*\|_p \leq \frac{p}{p-1} \|X_n\|_p.$$

Proof sketch. A corollary of the tail-sum formula for $\mathbb{E}((X_n^*)^p)$ together with the tail bound of Theorem 4.7. $\square$

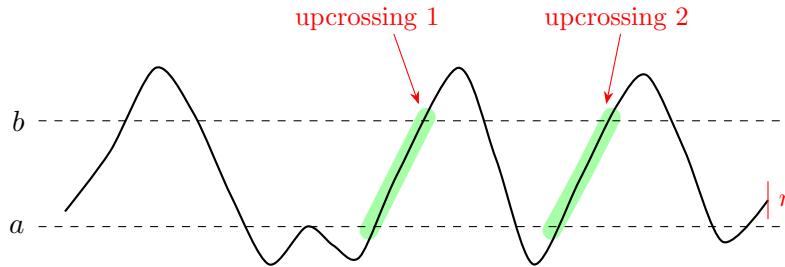
4.4 Towards convergence: upcrossings

Theorem 4.9 (Martingale convergence theorem). *Let $(X_n)_{n \geq 0}$ be an L^p -bounded submartingale. Then there exists $X_\infty \in L^p$ such that $X_n \rightarrow X_\infty$ a.s. Moreover, if*

1. $p = 1$ and $(X_n)_{n \geq 0}$ is uniformly integrable; or
2. $p > 1$ and $(X_n)_{n \geq 0}$ is a martingale or a nonnegative submartingale,

then $X_n \rightarrow X_\infty$ in L^p as well.

The mechanism behind the a.s. convergence is that a path that fails to converge must oscillate across some rational band $[a, b]$ infinitely often, and submartingales cannot afford to do that.



Lemma 4.10 (Upcrossing lemma). *Let $(X_n)_{n \geq 0}$ be a submartingale and let $U_n(a, b)$ denote the number of upcrossings of $[a, b]$ made by time n . Then*

$$(b - a) \mathbb{E}(U_n(a, b)) \leq \mathbb{E}((X_n - a)^+).$$

Lecture 5

Martingale Convergence

5.1 The convergence theorem

Lemma 5.1 (Upcrossing lemma). *Let $(X_n)_{n \geq 0}$ be a submartingale and let $U_n(a, b)$ be the number of upcrossings of $[a, b]$ made by time n . Then*

$$(b - a) \mathbb{E}(U_n(a, b)) \leq \mathbb{E}((X_n - a)^+).$$

Theorem 5.2 (Martingale convergence theorem). *Let $(X_n)_{n \geq 0}$ be an L^p -bounded submartingale. Then there exists $X_\infty \in L^p$ with $X_n \rightarrow X_\infty$ a.s. Moreover, if*

1. $p = 1$ and $(X_n)_{n \geq 0}$ is uniformly integrable; or
2. $p > 1$ and $(X_n)_{n \geq 0}$ is a martingale or a nonnegative submartingale,

then $X_n \rightarrow X_\infty$ in L^p .

Proof. Almost sure convergence. Let

$$U(a, b) := \lim_{n \rightarrow \infty} U_n(a, b),$$

which exists since $U_n(a, b)$ is nondecreasing in n . By monotone convergence,

$$\mathbb{E}(U(a, b)) = \lim_{n \rightarrow \infty} \mathbb{E}(U_n(a, b)).$$

Set

$$A_{ab} := \left\{ \omega : \liminf_{n \rightarrow \infty} X_n(\omega) < a < b < \limsup_{n \rightarrow \infty} X_n(\omega) \right\} = \{U(a, b) = \infty\}.$$

By Lemma 5.1 and $(x - a)^+ \leq |x| + |a|$,

$$(b - a) \mathbb{E}(U(a, b)) = (b - a) \lim_{n \rightarrow \infty} \mathbb{E}(U_n(a, b)) \leq \limsup_{n \rightarrow \infty} (|a| + \mathbb{E}(|X_n|)) < \infty,$$

using L^p -boundedness. Hence $U(a, b) < \infty$ a.s., so $\mathbb{P}(A_{ab}) = 0$. A path fails to converge exactly when it oscillates across some rational band, so

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} X_n \text{ d.n.e.}\right) = \mathbb{P}\left(\bigcup_{\substack{a < b \\ a, b \in \mathbb{Q}}} A_{ab}\right) \leq \sum_{\substack{a, b \in \mathbb{Q} \\ a < b}} \mathbb{P}(A_{ab}) = 0.$$

By Fatou's lemma,

$$\mathbb{E}(|X_\infty|^p) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(|X_n|^p) < \infty,$$

so there is $X_\infty \in L^p$ with $X_n \rightarrow X_\infty$ a.s.

(i) ✓ — this is the standard theorem on uniform integrability: a.s. convergence plus uniform integrability gives L^1 convergence.

(ii) Note $|X_n|$ is a nonnegative submartingale. Write $X_n^* := \max_{k=0, \dots, n} |X_k|$ and $X_\infty^* := \lim_{n \rightarrow \infty} X_n^*$. By monotone convergence and Doob's L^p inequality (Theorem 4.8),

$$\|X_\infty^*\|_p = \lim_{n \rightarrow \infty} \|X_n^*\|_p \leq \sup_{n \geq 0} \frac{p}{p-1} \|X_n\|_p < \infty.$$

So $X_n \rightarrow X_\infty$ a.s. and $|X_n| \leq X_\infty^* \in L^p$; dominated convergence gives $X_n \rightarrow X_\infty$ in L^p . $\square$

5.2 Application: opinion dynamics on a graph

Example 5.3 (Voter model). Let $G = (V, E)$ be a connected graph in which each vertex holds a value in $\{0, 1\}$ at time n , denoted $S_n(v)$ for $v \in V$; think of $S_0(v)$ as the initial belief of vertex v . At each step, independently of all other vertices, vertex v

$$\begin{cases} \text{keeps its belief} & \text{w.p. } p \in (0, 1), \\ \text{adopts a uniformly random neighbor's belief} & \text{w.p. } 1 - p. \end{cases}$$

Set

$$X_n := \sum_{v \in V} \deg(v) S_n(v).$$

Then $(X_n)_{n \geq 0}$ is a martingale — the degree weighting is exactly what makes the "adopt a neighbor" step mean-preserving.

Note $X_n \in \{0, 1, \dots, 2|E|\}$, so $(X_n)_{n \geq 0}$ is bounded and hence L^p -bounded for every p . By Theorem 5.2, $X_n \rightarrow X_\infty$, where X_∞ is supported on $\{0, 2|E|\}$ (the only absorbing configurations are total consensus):

$$X_\infty = \begin{cases} 0 & \text{w.p. } P_0, \\ 2|E| & \text{w.p. } 1 - P_0. \end{cases}$$

Let τ be the time at which beliefs polarize — a stopping time. Then

$$\sum_{v \in V} \deg(v) S_0(v) = \mathbb{E}(X_0) = \mathbb{E}(X_{n \wedge \tau}) \longrightarrow \mathbb{E}(X_\tau) = P_0 \cdot 0 + (1 - P_0) \cdot 2|E|,$$

so

$$\boxed{\frac{1}{2|E|} \sum_{v \in V} \deg(v) S_0(v) = 1 - P_0.}$$

The probability of reaching consensus on 1 is the degree-weighted fraction of vertices that started at 1.

5.3 Continuous-time martingales: definitions

Definition 5.4 (L^p -martingale). A process $(X_t)_{t \geq 0} \subset L^p$ is an L^p -martingale with respect to $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ if

1. X is adapted;
2. $\mathbb{E}(X_t \mid \mathcal{F}_s) = X_s$ for all $t > s \geq 0$.

It is a *submartingale* if (2) holds with $\geq$. It is L^p -bounded if $\sup_{t \geq 0} \mathbb{E}(|X_t|^p) < \infty$.

Definition 5.5 (Continuity). A (sub)martingale $(X_t)_{t \geq 0}$ is (right) continuous if

- (i) $\mathbb{F}$ is a standard filtration; and
- (ii) $t \mapsto X_t$ is (right) continuous, a.s.

Theorem 5.6 (Doob's regularization theorem). Let $(X_t)_{t \geq 0}$ be a submartingale with respect to a standard filtration $\mathbb{F}$. If $t \mapsto \mathbb{E}(X_t)$ is continuous, then there exists a version $(X'_t)_{t \geq 0}$ of X with:

1. X' is a submartingale;
2. all paths $t \mapsto X'_t(\omega)$ are càdlàg.

Notation. X' is a version of X if $\mathbb{P}\{X'_t = X_t\} = 1$ for every t . Càdlàg means right continuous with left limits.

Lecture 6

Continuous-Time Martingales

6.1 Versions

Definition 6.1. Processes $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$ are *versions* of one another if $\mathbb{P}(X_t = Y_t) = 1$ for every $t \geq 0$.

Remark 6.2. Doob's regularization theorem (Theorem 5.6) says every continuous-time martingale has a right continuous version. "Versions" will matter again for the Itô integral, which is only ever defined up to a version.

The point of the definition is that it is a statement at each *fixed* t , and therefore says nothing at all about the paths.

Example 6.3. Let $(B_t)_{t \geq 0}$ be a standard Brownian motion with continuous paths, independent of $U \sim \text{Unif}[0, 1]$, and set

$$B'_t = B_t + \mathbf{1}_{\{U=t\}}.$$

Then B' is a version of B : for each fixed t ,

$$\mathbb{P}(B'_t \neq B_t) = \mathbb{P}(U = t) = 0.$$

But every path of B' has a jump at $t = U(\omega)$, so B' has no continuous paths at all. Fixed-time agreement is much weaker than pathwise agreement.

6.2 Stopping preserves the martingale property

Theorem 6.4. Let $(X_t)_{t \geq 0}$ be an L^p (right) continuous martingale and let τ be a stopping time. Then $(X_{t \wedge \tau})_{t \geq 0}$ is also an L^p (right) continuous martingale.

Proof. Continuity of the stopped process comes for free from the construction below; the work is in adaptedness, integrability, and the martingale property.

Discretization. Let

$$T_n = \{k2^{-n} : k \in \mathbb{Z}_+ \cup \{0\}\}.$$

For $x \in \mathbb{R}$ put $x_n := 2^{-n} \lceil x 2^n \rceil \in T_n$, so that $x \leq x_n < x + 2^{-n}$ and $x_n \downarrow x$. Define $\tau_n(\omega) := 2^{-n} \lceil \tau(\omega) 2^n \rceil \in T_n$. This is a stopping time:

$$\{\tau_n \leq t\} = \{\lceil 2^n \tau \rceil \leq 2^n t\} = \{\lceil 2^n \tau \rceil \leq \lfloor 2^n t \rfloor\} = \left\{ \tau \leq \underbrace{2^{-n} \lfloor 2^n t \rfloor}_{\leq t} \right\} \in \mathcal{F}_t,$$

the middle step because $\lceil 2^n \tau \rceil$ is an integer.

Adaptedness. $(X_\Delta)_{\Delta \in T_n}$ is a discrete-time martingale, hence so is $(X_{\Delta \wedge \tau_n})_{\Delta \in T_n}$. For $s > t$ we have $t_n \wedge \tau_n < s$ for all n sufficiently large, so by right continuity

$$X_{t \wedge \tau} = \lim_{n \rightarrow \infty} X_{t_n \wedge \tau_n} \quad \text{a.s.}$$

is $\mathcal{F}_s$ -measurable for every $s > t$. Completeness of the filtration then gives that $X_{t \wedge \tau}$ is $\bigcap_{s>t} \mathcal{F}_s = \mathcal{F}_t$ -measurable, so $(X_{t \wedge \tau})_{t \geq 0}$ is adapted.

Integrability. Fix $k \in \mathbb{N}$ with $k > 1 + t \geq t_n \wedge \tau_n$ for all n . Then

$$\mathbb{E}(|X_{t \wedge \tau}|^p) \stackrel{\text{Fatou}}{\leq} \liminf_{n \rightarrow \infty} \mathbb{E}(|X_{t_n \wedge \tau_n}|^p) \stackrel{\text{Doob}}{\leq} \mathbb{E}(|X_k|^p), \quad (\star\star)$$

the second step by Doob's stopping theorem applied to the nonnegative submartingale $|X|^p$. Hence $(X_{t \wedge \tau})_{t \geq 0} \subset L^p$.

Martingale property. We have

$$\mathbb{E}(X_{t_n \wedge \tau_n} \mid \mathcal{F}_{s_n}) = X_{s_n \wedge \tau_n} \quad \forall n \geq 1,$$

and $X_{s_n \wedge \tau_n} \rightarrow X_{s \wedge \tau}$ a.s. Take any $F \in \mathcal{F}_s \subset \mathcal{F}_{s_n}$. Then

$$\mathbb{E}(X_{s_n \wedge \tau_n} \mathbf{1}_F) = \mathbb{E}(\mathbb{E}(X_{t_n \wedge \tau_n} \mid \mathcal{F}_{s_n}) \mathbf{1}_F) = \mathbb{E}(X_{t_n \wedge \tau_n} \mathbf{1}_F).$$

Now $X_{s_n \wedge \tau_n} \rightarrow X_{s \wedge \tau}$ and $X_{t_n \wedge \tau_n} \rightarrow X_{t \wedge \tau}$ a.s. and also in L^1 , so we may pass to the limit:

$$\mathbb{E}(\underbrace{X_{s \wedge \tau}}_{\mathcal{F}_s\text{-mbl.}} \mathbf{1}_F) = \mathbb{E}(X_{t \wedge \tau} \mathbf{1}_F) \implies X_{s \wedge \tau} = \mathbb{E}(X_{t \wedge \tau} \mid \mathcal{F}_s).$$

What remains is the L^1 convergence, for which we need $(X_{t_n \wedge \tau_n})_{n \geq 1}$ uniformly integrable for each t . $\square$

Theorem 6.5 (de la Vallée Poussin). *A family $(X_\alpha)_{\alpha \in A}$ is uniformly integrable if and only if there exists a convex $\varphi: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ with $\varphi(x)/x \rightarrow \infty$ as $x \rightarrow \infty$ and*

$$\sup_{\alpha \in A} \mathbb{E}(\varphi(|X_\alpha|)) < \infty.$$

Taking $\varphi(x) = x^p$ for $p > 1$ — convex, with $\varphi(x)/x = x^{p-1} \rightarrow \infty$ — the uniform bound $(\star\star)$ is exactly the hypothesis of Theorem 6.5, so $(X_{t \wedge \tau})$ is uniformly integrable and the proof of Theorem 6.4 is complete.

Remark 6.6. This is where $p > 1$ is used. For $p = 1$ the bound $(\star\star)$ gives no uniform integrability on its own, and the hypothesis has to be assumed separately.

Lecture 7

Maximal Inequalities and Local Martingales

7.1 Recap

(Right) continuous martingales are stable after stopping: if $(X_t)_{t \geq 0}$ is an L^p martingale with (right) continuous paths a.s. and τ is a stopping time, then $(X_{t \wedge \tau})_{t \geq 0}$ is also an L^p martingale with (right) continuous paths a.s.

The proof is a good example of the standard move: discretize, apply the discrete-time martingale results, then recover the continuous-time statement using continuity of the paths and regularity of the underlying filtration. We use it again below.

7.2 Doob's inequalities in continuous time

Theorem 7.1. *Let $(X_t)_{t \geq 0}$ be a nonnegative right continuous L^p submartingale, and write*

$$X_T^* = \sup_{t \in [0, T]} |X_t|.$$

Then

$$\lambda \mathbb{P}\{X_T^* \geq \lambda\} \leq \mathbb{E}(X_T) \quad \forall T \geq 0, \lambda > 0,$$

and

$$\|X_T^*\|_p \leq \frac{p}{p-1} \|X_T\|_p \quad \forall p > 1.$$

Proof. Take $T_n = \{kT2^{-n} : k \in \{0, \dots, 2^n\}\}$, a discretization of $[0, T]$. Then $(X_\Delta)_{\Delta \in T_n}$ is a discrete-time submartingale, and by right continuity $\max_{\Delta \in T_n} X_\Delta \uparrow X_T^*$. So by Fatou and the discrete-time inequality (Theorem 4.8),

$$\|X_T^*\|_p \leq \liminf_{n \rightarrow \infty} \left\| \max_{\Delta \in T_n} X_\Delta \right\|_p \leq \frac{p}{p-1} \|X_T\|_p.$$

□

Example 7.2 (Tail bound for Brownian motion). Let $(B_t)_{t \geq 0}$ be a Brownian motion. Then

$$\mathbb{P} \left\{ \sup_{t \in [0,1]} B_t \geq \lambda \right\} \leq e^{-\lambda^2/2}, \quad \lambda \geq 0.$$

Indeed, $e^{\alpha B_t}$ is a nonnegative submartingale for every $\alpha \in \mathbb{R}$ (it is a martingale after dividing by $e^{\alpha^2 t/2}$). For $\alpha > 0$,

$$\mathbb{P} \left\{ \sup_{t \in [0,1]} B_t \geq \lambda \right\} = \mathbb{P} \left\{ \sup_{t \in [0,1]} e^{\alpha B_t} \geq e^{\alpha \lambda} \right\} \leq e^{-\alpha \lambda} \mathbb{E}(e^{\alpha B_1}) = e^{-\alpha \lambda} e^{\alpha^2/2},$$

by the first inequality of Theorem 7.1. Optimizing,

$$\inf_{\alpha > 0} e^{-\alpha \lambda} e^{\alpha^2/2} = e^{-\lambda^2/2}$$

at $\alpha = \lambda$.

7.3 Convergence in continuous time

Theorem 7.3. Let $(X_t)_{t \geq 0}$ be an L^p -bounded right continuous submartingale. Then there exists $X_\infty \in L^p$ with $X_t \rightarrow X_\infty$ a.s. Moreover if

1. $p = 1$ and $(X_t)_{t \geq 0}$ is uniformly integrable; or
2. $p > 1$ and $(X_t)_{t \geq 0}$ is a martingale or a nonnegative submartingale,

then $X_t \rightarrow X_\infty$ in L^p also.

7.4 Examples of continuous-time martingales

Example 7.4.

- $(B_t)_{t \geq 0}$, a standard Brownian motion — continuous sample paths.
- $(N_t - \lambda t)_{t \geq 0}$ with $(N_t)_{t \geq 0} \sim \text{PP}(\lambda)$ — discontinuous sample paths (càdlàg). This is the *compensated* Poisson process.
- $(B_t^2 - t)_{t \geq 0}$ and $(\exp(\alpha B_t - \alpha^2 t/2))_{t \geq 0}$ are martingales.

From $(B_t^2 - t)_{t \geq 0}$: if τ is a stopping time with $\mathbb{E}(\tau) < \infty$, then optional stopping gives $\mathbb{E}(B_\tau^2) = \mathbb{E}(\tau)$, and since $\mathbb{E}(B_\tau) = 0$ this reads

$$\mathbb{E}(\tau) = \text{Var}(B_\tau).$$

Example 7.5 (A continuous martingale that is not uniformly integrable). Let $(B_t)_{t \geq 0}$ be a standard Brownian motion and $\tau = \inf\{t \geq 0 : B_t = 1\}$, which is finite a.s. Define

$$X_t = \begin{cases} B_{\frac{t}{1-t} \wedge \tau} & 0 \leq t < 1, \\ 1 & t \geq 1. \end{cases}$$

Then

$$\mathbb{E}(X_t) = \begin{cases} 0 & \text{for } t \in [0, 1), \\ 1 & t \geq 1. \end{cases}$$

On $[0, 1)$ this is a martingale (a deterministic time change of the stopped Brownian motion $B_{\cdot \wedge \tau}$), and $X_t \rightarrow B_\tau = 1$ a.s. as $t \uparrow 1$, so the paths are in fact continuous. What fails is uniform integrability: the a.s. limit does not carry the expectation, and $t \mapsto \mathbb{E}(X_t)$ jumps at $t = 1$. In particular the hypothesis of Doob's regularization theorem is not satisfied.

7.5 Local martingales

Definition 7.6 (Local martingale). A process $(X_t)_{t \geq 0}$ is a *local martingale* if

1. X_0 is $\mathcal{F}_0$ -measurable;
2. there is a nondecreasing sequence of stopping times $(\tau_n)_{n \geq 1}$ with $\tau_n \rightarrow \infty$ a.s.; and
3. $(X_{t \wedge \tau_n} - X_0)_{t \geq 0}$ is a martingale for every $n \geq 1$.

The sequence $(\tau_n)_{n \geq 1}$ is a *localizing sequence* for X .

Example 7.5 is the reason this notion is needed: a process can behave like a martingale on every bounded piece of its life without being a martingale globally, and stopping is the right way to isolate the good behavior.

Lecture 8

Predictable Processes and the Itô Integral

The Itô integral is the continuous-time version of the martingale transform. Its integrands are the *predictable* processes, the continuous-time analogue of previsible processes.

8.1 The predictable σ -algebra

Definition 8.1 (Predictable rectangle). A *predictable rectangle* is a set $R \subset \mathbb{R}^+ \times \Omega$ of the form

$$R = \begin{cases} \{0\} \times F & \text{for } F \in \mathcal{F}_0, \\ (s, t] \times F & \text{for } F \in \mathcal{F}_s, 0 \leq s \leq t. \end{cases}$$

Write $\mathcal{R} := \{\text{all such } R\} \subset 2^{\mathbb{R}^+ \times \Omega}$, and

$$\mathcal{P} := \sigma(\mathcal{R}) = \text{the smallest } \sigma\text{-algebra containing every } R,$$

the *predictable σ -algebra*.

Definition 8.2 (Predictable process). A process $H: (t, \omega) \mapsto H_t(\omega)$ is *predictable* if H is $\mathcal{P}$ -measurable, i.e.

$$\{(t, \omega) : H_t(\omega) \leq \alpha\} \in \mathcal{P} \quad \forall \alpha \in \mathbb{R}.$$

Example 8.3. Take $H = \mathbf{1}_R$ with $R \in \mathcal{R}$, so

$$H_t(\omega) = \mathbf{1}_{(s_1, s_2]}(t) \mathbf{1}_F(\omega), \quad F \in \mathcal{F}_{s_1}.$$

At time s_1 we can already predict the entire future of this process with certainty. That is the content of predictability: given $\sigma(\bigcup_{s < t} \mathcal{F}_s)$ we can determine H_t .

Definition 8.4 (Simple predictable process). The *simple predictable* processes are the linear combinations of indicators of predictable rectangles:

$$H_t = A_0 \mathbf{1}_{\{0\}}(t) + \sum_{i=1}^k A_i \mathbf{1}_{(u_i, v_i]}(t), \quad t \geq 0,$$

where $u_i < v_i$ for $i = 1, \dots, k$, each A_i is a simple $\mathcal{F}_{u_i}$ -measurable random variable, and A_0 is a simple $\mathcal{F}_0$ -measurable random variable.

Such an H is $\mathcal{P}$ -measurable by construction, hence predictable — and so is every pointwise limit of such processes.

Remark 8.5 (Equivalent description). $\mathcal{P}$ is the σ -algebra on $\mathbb{R}^+ \times \Omega$ generated by all left continuous adapted processes. In particular, every left continuous adapted process is predictable.

Example 8.6. If H is predictable and τ is a stopping time, then $(\mathbf{1}_{\{t \leq \tau\}} H_t)_{t \geq 0}$ is predictable.

8.2 Setting

Setting. Work on $(\Omega, \mathcal{F}, \mathbb{P})$ with a standard filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$, and let $(B_t)_{t \geq 0}$ be a standard $\mathbb{F}$ -Brownian motion, meaning

- $B_0 = 0$ and $\mathbb{E}(B_1^2) = 1$;
- $(B_t)_{t \geq 0}$ is adapted to $\mathbb{F}$;
- $B_t - B_s$ is independent of $\mathcal{F}_s$ for all $0 \leq s < t$.

Goal. For predictable H , define and understand the properties of the Itô integral

$$\int H dB.$$

Notation. The symbol $\int$ here is *just notation*. It is not a Riemann integral, and none of the usual constructions of one apply.

8.3 Construction on simple integrands

Step 1: predictable rectangles

For $R \in \mathcal{R}$, define

$$\int \mathbf{1}_R dB := \begin{cases} 0 & \text{if } R = \{0\} \times F, F \in \mathcal{F}_0, \\ \mathbf{1}_F(B_t - B_s) & \text{if } R = (s, t] \times F, F \in \mathcal{F}_s. \end{cases}$$

Remark 8.7 (Why this is the right definition). The following manipulation is purely formal — it is a mnemonic, not an argument, since the objects it moves around are not yet defined:

$$\begin{aligned} \left(\int \mathbf{1}_R dB \right) (\omega) &= \int \mathbf{1}_{(s, t]}(u) \mathbf{1}_F(\omega) dB_u(\omega) = \int_s^t \mathbf{1}_F(\omega) dB_u(\omega) \\ &= \mathbf{1}_F(\omega) \int_s^t dB_u(\omega) = \mathbf{1}_F(\omega) (B_t - B_s)(\omega). \end{aligned}$$

It lands on the definition above, which is what we take as given.

Step 2: simple predictable integrands

Write

$$H_t = A_0 \mathbf{1}_{\{0\}}(t) + \sum_{i=1}^k A_i \mathbf{1}_{(u_i, v_i]}(t).$$

Extending by linearity,

$$\int H dB = \sum_{i=1}^k A_i (B_{v_i} - B_{u_i}),$$

which is exactly a *martingale transform*: an $\mathcal{F}_{u_i}$ -measurable weight applied to the increment that follows it. The A_0 term contributes nothing, since the single time $\{0\}$ carries no increment.

Lecture 9

The Itô Isometry

Recall: for a simple predictable process

$$H_t = A_0 \mathbf{1}_{\{0\}}(t) + \sum_{i=1}^k A_i \mathbf{1}_{(u_i, v_i]}(t), \quad A_i \text{ a simple } \mathcal{F}_{u_i}\text{-measurable random variable,}$$

the integral

$$\int H dB := \sum_{i=1}^k A_i (B_{v_i} - B_{u_i})$$

is well defined.

9.1 The isometry

Theorem 9.1 (Itô isometry). *For simple predictable H ,*

$$\mathbb{E} \left(\left(\int H dB \right)^2 \right) = \mathbb{E} \left(\int_{\mathbb{R}^+} H_t^2 dt \right).$$

Read this as an equality of norms,

$$\underbrace{\left\| \int H dB \right\|_{L^2(\Omega, \mathcal{F}, \mathbb{P})}^2}_{\text{a random variable}} = \underbrace{\|H\|_{L^2(\mathbb{R}^+ \times \Omega, \mathcal{P}, \mathbb{P} \times \text{Leb})}^2}_{\text{a process}}$$

by analogy with the elementary identity $\int_{\mathbb{R}} f^2(t) dt = \|f\|_{L^2(\mathbb{R}, \mathcal{B}, \text{Leb})}^2$. So $H \mapsto \int H dB$ is a transformation from predictable processes to L^2 random variables that preserves norms — much like Parseval's theorem for the Fourier transform. That is exactly the property that will let us extend it by continuity.

Proof. Without loss of generality the intervals $(u_i, v_i]$ are disjoint, relabelled so that

$$u_1 < v_1 \leq u_2 < v_2 \leq \cdots \leq u_k < v_k.$$

Abbreviate $\Delta_i B := B_{v_i} - B_{u_i}$. Then

$$\begin{aligned} \mathbb{E} \left(\left(\int H dB \right)^2 \right) &= \mathbb{E} \left(\sum_{i=1}^k \sum_{j=1}^k A_i A_j \Delta_i B \Delta_j B \right) = \sum_{i=1}^k \sum_{j=1}^k \mathbb{E} (A_i A_j \Delta_i B \Delta_j B) \\ &= \sum_{i=1}^k \sum_{\substack{j=1 \\ j \neq i}}^k \underbrace{\mathbb{E} (A_i A_j \Delta_i B) \mathbb{E} (\Delta_j B)}_{=0} + \sum_{i=1}^k \mathbb{E} (A_i^2 (\Delta_i B)^2) \\ &= \sum_{i=1}^k \mathbb{E} (A_i^2) (v_i - u_i) = \mathbb{E} \left(\int_{\mathbb{R}^+} H_t^2 dt \right). \end{aligned}$$

For the cross terms, take $i < j$: then $A_i A_j \Delta_i B$ is $\mathcal{F}_{u_j}$ -measurable, while $\Delta_j B$ is independent of $\mathcal{F}_{u_j}$ with mean zero, so the expectation factors and vanishes. For the diagonal terms, A_i^2 is $\mathcal{F}_{u_i}$ -measurable and $(\Delta_i B)^2$ is independent of $\mathcal{F}_{u_i}$ with mean $v_i - u_i$. The last equality is just $\int_{\mathbb{R}^+} H_t^2 dt = \sum_i A_i^2 (v_i - u_i)$, using disjointness. $\square$

9.2 Extension to L^2 integrands

Definition 9.2.

$$\mathcal{L}_0^2 := \left\{ H \text{ predictable} : \mathbb{E} \left(\int_{\mathbb{R}^+} H_t^2 dt \right) < \infty \right\}.$$

Take $H \in \mathcal{L}_0^2$. It can be approximated in the $\mathcal{L}_0^2$ norm by a sequence of simple predictable processes $(H^{(n)})_{n \geq 1}$, i.e.

$$\mathbb{E} \left(\int_{\mathbb{R}^+} (H_t^{(n)} - H_t)^2 dt \right) \longrightarrow 0.$$

Then $(H^{(n)})_{n \geq 1}$ is Cauchy in $\mathcal{L}_0^2$, so by Theorem 9.1 the random variables $\int H^{(n)} dB$ are Cauchy in $L^2(\Omega, \mathcal{F}, \mathbb{P})$. Completeness gives a limit: there is a random variable in L^2 , which we call $\int H dB$, with

$$\int H^{(n)} dB \longrightarrow \int H dB \quad \text{in } L^2.$$

The limit does not depend on the approximating sequence, so this is well defined.

9.3 The Itô integral as a process

So far $H \mapsto \int H dB$ is a map from processes in $\mathcal{L}_0^2$ to random variables in L^2 . To get a *process*, introduce

$$\mathbf{1}_{[0,t]} H : (s, \omega) \mapsto \mathbf{1}_{[0,t]}(s) H_s(\omega)$$

and set

$$X_t := \int \mathbf{1}_{[0,t]} H dB. \quad (\star)$$

(Whether one writes $\mathbf{1}_{[0,t]}$ or $\mathbf{1}_{(0,t]}$ is immaterial: the slice at $t = 0$ integrates to 0 by definition.)

This is in fact well defined for every H in the larger class

$$\mathcal{L}^2 := \left\{ H \text{ predictable} : \mathbb{E} \int_0^t H_s^2 ds < \infty \quad \forall t \geq 0 \right\},$$

since $\mathbf{1}_{[0,t]}H \in \mathcal{L}_0^2$ for each fixed t . Observe that $(X_t)_{t \geq 0} \subset L^2$, by the Itô isometry.

Q. What other properties does X have?

Theorem 9.3. *For every $H \in \mathcal{L}^2$, the process $(X_t)_{t \geq 0}$ defined by $(\star)$ is an L^2 -martingale. Moreover there is a version all of whose paths are continuous.*

Definition 9.4. For $H \in \mathcal{L}^2$, write

$$\left(\int_0^t H_s dB_s \right)_{t \geq 0}$$

for the continuous L^2 -martingale satisfying $\int_0^t H_s dB_s = \int \mathbf{1}_{[0,t]} H dB$ a.s. for all $t \geq 0$. Also define, for $r < t$,

$$\int_r^t H_s dB_s := \int_0^t H_s dB_s - \int_0^r H_s dB_s = \int \mathbf{1}_{(r,t]} H dB.$$

Proof of Theorem 9.3, martingale property. Fix $T > 0$ and take simple predictable $(H^{(n)})_{n \geq 1}$ with

$$\mathbf{1}_{[0,T]} H^{(n)} \longrightarrow \mathbf{1}_{[0,T]} H \quad \text{in } \mathcal{L}_0^2.$$

Writing $H^{(n)} = \sum_i A_i \mathbf{1}_{(s_i, s_{i+1}]}$,

$$X_t^{(n)} := \int \mathbf{1}_{[0,t]} H^{(n)} dB = \sum_i A_i (B_{s_{i+1} \wedge t} - B_{s_i \wedge t}),$$

which is a martingale in t — it is a martingale transform of the Brownian increments. By the Itô isometry, $X_t^{(n)} \rightarrow X_t$ in L^2 for every $t \in [0, T]$. An L^2 limit of L^2 -martingales is an L^2 -martingale (HW5 Q1), so $(X_t)_{t \in [0, T]}$ is an L^2 -martingale, and since T was arbitrary so is $(X_t)_{t \geq 0}$. $\square$

The existence of a continuous version is the more delicate half; we prove it next time, in Theorem 10.1.

Lecture 10

Properties of the Itô Integral

Last time: the Itô isometry, and the extension of the Itô integral from simple predictable integrands to all of $\mathcal{L}^2$.

10.1 The Itô integral is a continuous martingale

Theorem 10.1 (restating Theorem 9.3). *For every $H \in \mathcal{L}^2$, the process $(X_t)_{t \geq 0}$ defined by*

$$X_t := \int \mathbf{1}_{[0,t]} H dB$$

is a martingale. Moreover it admits a continuous version, denoted

$$\left(\int_0^t H_s dB_s \right)_{t \geq 0}.$$

Proof. Fix $T > 0$ and take simple predictable $(H^{(n)})_{n \geq 1}$ with

$$\mathbf{1}_{[0,T]} H^{(n)} \longrightarrow \mathbf{1}_{[0,T]} H \quad \text{in } \mathcal{L}^2.$$

By the Itô isometry,

$$X_t^{(n)} := \int \mathbf{1}_{[0,t]} H^{(n)} dB \longrightarrow X_t \quad \text{in } L^2, \quad \forall t \in [0, T],$$

and each $X^{(n)}$ is a martingale (the integral of a simple predictable integrand is). An L^2 limit of martingales is a martingale, so $(X_t)_{t \geq 0}$ is a martingale.

For continuity: since $X_T^{(n)} \rightarrow X_T$ in L^2 , the sequence $(X_T^{(n)})_{n \geq 1}$ is Cauchy in L^2 , so we may extract a subsequence $(n_k)_{k \geq 1}$ with

$$\mathbb{E} \left(|X_T^{(n_{k+1})} - X_T^{(n_k)}|^2 \right) \leq 2^{-3k} \quad \forall k \geq 1.$$

The difference $X^{(n_{k+1})} - X^{(n_k)}$ is again a continuous martingale, so Doob's L^2 maximal inequality applies:

$$\mathbb{P} \left(\sup_{t \in [0, T]} |X_t^{(n_{k+1})} - X_t^{(n_k)}| > 2^{-k} \right) \leq 2^{2k} \mathbb{E} \left(|X_T^{(n_{k+1})} - X_T^{(n_k)}|^2 \right) \leq 2^{-k}.$$

These probabilities are summable, so by Borel–Cantelli

$$\mathbb{P} \left(\sup_{t \in [0, T]} |X_t^{(n_{k+1})} - X_t^{(n_k)}| > 2^{-k} \text{ i.o.} \right) = 0.$$

Hence with probability 1 the sequence $(X^{(n_k)})_{k \geq 1}$ is uniformly Cauchy on $[0, T]$, so

$$X'_t := \lim_{k \rightarrow \infty} X_t^{(n_k)}$$

exists a.s., and w.p. 1, $\sup_{t \in [0, T]} |X_t^{(n_k)} - X'_t| \rightarrow 0$. A uniform limit of continuous functions is continuous, so X' is continuous. Finally $X_t^{(n_k)} \rightarrow X'_t$ in L^2 for each fixed t , so $X'_t = X_t$ a.s. for every $t \in [0, T]$: X' is the desired continuous version. $\square$

Remark 10.2. Continuity is what lets us write $\int_0^t H_s dB_s$ as a genuine process and evaluate it at random times. Both of the properties below use it.

10.2 Pulling out an $\mathcal{F}_s$ -measurable factor

Theorem 10.3. *Let $H \in \mathcal{L}^2$, let $0 \leq s \leq t$, and let Z be bounded and $\mathcal{F}_s$ -measurable. Then*

$$\mathbf{1}_{(s, t]} Z : (u, \omega) \mapsto \mathbf{1}_{(s, t]}(u) Z(\omega)$$

is predictable, and

$$Z \int_s^t H_u dB_u = \int \mathbf{1}_{(s, t]} Z H dB \quad a.s.$$

Proof. Choose a sequence of $\mathcal{F}_s$ -measurable *simple* random variables $(Z_n)_{n \geq 1}$ with $|Z_n| \leq |Z|$ and $Z_n \rightarrow Z$ a.s. Then

$$\underbrace{\mathbf{1}_{(s, t]} Z_n}_{\text{simple predictable}} \longrightarrow \mathbf{1}_{(s, t]} Z \quad a.s.,$$

so $\mathbf{1}_{(s, t]} Z$ is predictable, and hence so is $\mathbf{1}_{(s, t]} Z H$; boundedness of Z gives $\mathbf{1}_{(s, t]} Z H \in \mathcal{L}^2$.

Let $(H_n)_{n \geq 1}$ be simple predictable with $\mathbf{1}_{(0, t]} H_n \rightarrow \mathbf{1}_{(0, t]} H$ in $\mathcal{L}^2$. Decompose

$$\begin{aligned} \int \mathbf{1}_{(s, t]} Z H dB - Z \int_s^t H dB &= \underbrace{\int \mathbf{1}_{(s, t]} (Z H - Z_n H_n) dB}_{\text{(A)}} \\ &\quad + \underbrace{\left(\int \mathbf{1}_{(s, t]} Z_n H_n dB - Z_n \int \mathbf{1}_{(s, t]} H_n dB \right)}_{\text{(B)}} \\ &\quad + \underbrace{Z_n \int \mathbf{1}_{(s, t]} (H_n - H) dB}_{\text{(C)}} + \underbrace{(Z_n - Z) \int \mathbf{1}_{(s, t]} H dB}_{\text{(D)}}. \end{aligned}$$

Now, term by term:

- (A) $\rightarrow 0$ in L^2 , by the Itô isometry and dominated convergence.
- (B) $= 0$, by the definition of the Itô integral for simple integrands — for a simple integrand an $\mathcal{F}_s$ -measurable constant really does factor out of each summand.
- (C) $\rightarrow 0$ in L^2 , by the Itô isometry and boundedness of Z_n .
- (D) $\rightarrow 0$, by dominated convergence.

The left-hand side does not depend on n , so it is 0 a.s. □

10.3 Stopping the Itô integral

Theorem 10.4. *Let $H \in \mathcal{L}^2$ and let τ be a stopping time. Then*

$$\int_0^{t \wedge \tau} H_s dB_s = \int \mathbf{1}_{[0, t \wedge \tau]} H dB \quad a.s.$$

Proof. Write $X_t := \int_0^t H_s dB_s$, which is a martingale by Theorem 10.1. The claim is equivalent to

$$X_\tau = \int \mathbf{1}_{(0, \tau]} H dB \quad a.s. \text{ for every bounded stopping time } \tau,$$

since $t \wedge \tau$ is a bounded stopping time for each fixed t .

Suppose $\tau < m$, and define the dyadic grid

$$\mathcal{T}_n = \{k2^{-n} : k = 0, \dots, m2^n\}.$$

Then $\tau_n := 2^{-n} \lceil 2^n \tau \rceil$ is a stopping time taking values in $\mathcal{T}_n$, and $\tau_n \downarrow \tau$. Its graph decomposes as

$$\{(t, \omega) \in \mathbb{R}^+ \times \Omega : t \leq \tau_n(\omega)\} = (\{0\} \times \Omega) \cup \bigcup_{k=0}^{m2^n} (k2^{-n}, (k+1)2^{-n}] \times \{\tau > k2^{-n}\},$$

using that $\tau_n \geq (k+1)2^{-n} \iff \tau > k2^{-n}$. Telescoping and then applying this decomposition,

$$\begin{aligned} X_{\tau_n} &= \sum_{k=0}^{m2^n} \mathbf{1}_{\{\tau_n \geq (k+1)2^{-n}\}} (X_{(k+1)2^{-n}} - X_{k2^{-n}}) \\ &= \sum_{k=0}^{m2^n} \mathbf{1}_{\{\tau > k2^{-n}\}} \int \mathbf{1}_{(k2^{-n}, (k+1)2^{-n}]} H dB \\ &= \int \sum_{k=0}^{m2^n} \mathbf{1}_{\{\tau > k2^{-n}\} \times (k2^{-n}, (k+1)2^{-n}]} H dB \\ &= \int \mathbf{1}_{[0, \tau_n]} H dB = \int_0^{\tau_n} H_s dB_s. \end{aligned}$$

The second equality is Theorem 10.3, applied with the bounded $\mathcal{F}_{k2^{-n}}$ -measurable random variable $Z = \mathbf{1}_{\{\tau > k2^{-n}\}}$.

Since $\tau_n \downarrow \tau$ and X has continuous paths, $X_{\tau_n} \rightarrow X_\tau$ as $n \rightarrow \infty$, which gives the claim. □

Lecture 11

Localization: the Itô Integral on $\mathcal{L}_{\text{loc}}^2$

11.1 Where we stand

For $H \in \mathcal{L}^2$,

$$\int_0^t H_s dB_s = \int \mathbf{1}_{[0,t]} H dB$$

is a continuous martingale, and it satisfies the properties one would want:

1. Z bounded and $\mathcal{F}_s$ -measurable $\implies Z \int_s^t H_u dB_u = \int_s^t Z H_u dB_u$;
2. τ a stopping time $\implies \int_0^{t \wedge \tau} H_u dB_u = \int \mathbf{1}_{[0, t \wedge \tau]} H dB$ a.s.

So the “Riemann” impulse to manipulate the notation is largely justified. But the Itô integral is doing something genuinely new, beyond Riemann integration.

Example 11.1. Let $b: \mathbb{R}^+ \rightarrow \mathbb{R}$ be smooth with $b(0) = 0$. Ordinary calculus gives

$$2 \int_0^t b(s) db(s) = b^2(t) - b^2(0) = b^2(t).$$

Contrast this with the Brownian case, where

$$2 \int_0^t B_s dB_s = B_t^2 - t \quad \forall t \geq 0.$$

The extra $-t$ is the whole story of Itô’s lemma: put in any integrand and it splits off a martingale plus a correction. We will prove this shortly.

11.2 The obstruction: transformed integrands leave $\mathcal{L}^2$

Any good theory of calculus should allow smooth transformation of integrands. For a smooth path b ,

$$\int_u^v f'(b(t)) db(t) = \int_{b(u)}^{b(v)} f'(x) dx = f(b(v)) - f(b(u)).$$

So ideally we would like to allow integrands of the form

$$\int_0^t f'(B_s) dB_s.$$

The problem is that nice transformations of integrands in $\mathcal{L}^2$ are in general no longer in $\mathcal{L}^2$.

Example 11.2. Take $f(x) = e^{x^4}$, so the candidate integrand is $f(B_s) = e^{B_s^4}$. Then

$$\mathbb{E} \left(\int_0^t f(B_s)^2 ds \right) = \infty \quad \forall t > 0,$$

since $\mathbb{E}(e^{2B_s^4}) = \infty$ for every $s > 0$. So $f(B) \notin \mathcal{L}^2$ and the Itô isometry has nothing to say about it.

Q. How do we make sense of $\int_0^t e^{B_s^4} dB_s$?

The integrand fails an *integrability* condition, not a pathwise one: for each fixed ω the map $s \mapsto e^{B_s(\omega)^4}$ is perfectly nice on $[0, t]$. That suggests weakening the requirement from an expectation to an almost sure statement.

Definition 11.3 (Locally square integrable integrands).

$$\mathcal{L}_{\text{loc}}^2 := \left\{ H \text{ predictable} : \int_0^t H_s^2 ds < \infty \text{ a.s. } \forall t \geq 0 \right\}.$$

Definition 11.4 (Localizing sequence). Let $H \in \mathcal{L}_{\text{loc}}^2$. A *localizing sequence* for H is a nondecreasing sequence of stopping times $(\tau_k)_{k \geq 1}$ with $\tau_k \rightarrow \infty$ a.s. and

$$\mathbf{1}_{[0, \tau_k]} H \in \mathcal{L}^2 \quad \forall k \geq 1.$$

Example 11.5 (THE example). For $H \in \mathcal{L}_{\text{loc}}^2$, set

$$\tau_k := \inf \left\{ t \geq 0 : \int_0^t H_s^2 ds \geq k \right\}.$$

This is a localizing sequence: by construction $\int_0^{\tau_k} H_s^2 ds \leq k$, so $\mathbb{E} \int \mathbf{1}_{[0, \tau_k]} H_s^2 ds \leq k < \infty$, and $\tau_k \rightarrow \infty$ a.s. because $\int_0^t H_s^2 ds < \infty$ for every t .

11.3 The extension theorem

Theorem 11.6. For every $H \in \mathcal{L}_{\text{loc}}^2$ there exists a local martingale $(X_t)_{t \geq 0}$ with continuous paths such that, for every localizing sequence $(\tau_k)_{k \geq 1}$ for H ,

$$X_{t \wedge \tau_k} = \int \mathbf{1}_{[0, t \wedge \tau_k]} H dB \quad \text{a.s.}$$

Moreover this process is unique up to versions.

So, as before, we introduce the notation $\int_0^t H_s dB_s$ to denote the continuous local martingale X at time t .

Proof. Existence. Fix a localizing sequence $(\tau_n)_{n \geq 1}$ and define the continuous martingales

$$X_t^{(n)} := \int_0^t \mathbf{1}_{[0, \tau_n]} H_s dB_s, \quad n \geq 1,$$

which makes sense because $\mathbf{1}_{[0, \tau_n]} H \in \mathcal{L}^2$. By Theorem 10.4 and $\tau_n \wedge \tau_{n+1} = \tau_n$,

$$X_{t \wedge \tau_n}^{(n+1)} = \int_0^t \underbrace{\mathbf{1}_{[0, \tau_n \wedge \tau_{n+1}]}}_{= \tau_n} H_s dB_s = X_t^{(n)} \quad \text{a.s.}$$

Both sides are continuous in t , so the exceptional set can be taken independent of t :

$$\underbrace{\mathbb{P} \left\{ X_{t \wedge \tau_n}^{(n+1)} = X_t^{(n)} \quad \forall t \geq 0 \right\}}_{=: \Omega_n} = 1.$$

On $\bigcap_{n \geq 1} \Omega_n$ the processes agree consistently on the growing intervals $[0, \tau_n]$, and since $\tau_n \rightarrow \infty$ a.s. the limit

$$X_t := \lim_{n \rightarrow \infty} X_t^{(n)}$$

exists for every t and is continuous.

Uniqueness. Suppose $(Y_t)_{t \geq 0}$ and a localizing sequence $(\mu_n)_{n \geq 1}$ for H satisfy $Y_{t \wedge \mu_n} = \int \mathbf{1}_{[0, t \wedge \mu_n]} H dB$. Define

$$Y_t^{(n)} := \int \mathbf{1}_{[0, t \wedge \mu_n]} H dB,$$

a continuous martingale, so $Y_t^{(n)} = Y_{t \wedge \mu_n}$ a.s. Then

$$Y_{t \wedge \tau_n}^{(n)} = \int \mathbf{1}_{[0, t \wedge \tau_n \wedge \mu_n]} H dB = X_{t \wedge \mu_n}^{(n)} \quad \text{a.s.},$$

so

$$\mathbf{1}_{\{t \leq \mu_n \wedge \tau_n\}} Y_t^{(n)} = \mathbf{1}_{\{t \leq \mu_n \wedge \tau_n\}} X_t^{(n)} \quad \forall n \geq 1.$$

Letting $n \rightarrow \infty$ and using $\mu_n \wedge \tau_n \rightarrow \infty$ a.s.,

$$\mathbf{1}_{\{t \leq \mu_n \wedge \tau_n\}} Y_t = \mathbf{1}_{\{t \leq \mu_n \wedge \tau_n\}} X_t \quad \longrightarrow \quad Y_t = X_t \quad \text{a.s.}, \quad \forall t \geq 0.$$

So $(Y_t)_{t \geq 0}$ is a version of $(X_t)_{t \geq 0}$. □

11.4 Properties of the Itô integral on $\mathcal{L}_{\text{loc}}^2$

Collecting everything, for $G, H \in \mathcal{L}_{\text{loc}}^2$:

1. $\left(\int_0^t H_s dB_s \right)_{t \geq 0}$ is a continuous local martingale, and for any localizing sequence $(\tau_n)_{n \geq 1}$ for H ,

$$\int_0^{t \wedge \tau_n} H_s dB_s = \int \mathbf{1}_{[0, t \wedge \tau_n]} H dB \quad \text{a.s.} \quad \forall t \geq 0.$$

2. (Linearity) For $a, b \in \mathbb{R}$,

$$\int_0^t (aG_s + bH_s) dB_s = a \int_0^t G_s dB_s + b \int_0^t H_s dB_s.$$

3. (stopping time) For any stopping time τ ,

$$\int_0^{t \wedge \tau} H_s dB_s = \int_0^t \mathbf{1}_{[0, \tau]} H_s dB_s.$$

4. (Pulling out a factor) For Z bounded and $\mathcal{F}_s$ -measurable,

$$Z \int_s^t H_u dB_u = \int_s^t Z H_u dB_u.$$

Remark 11.7. The price of localization is that we lose the martingale property: on $\mathcal{L}_{\text{loc}}^2$ the Itô integral is only a *local* martingale, and only a genuine martingale under extra integrability. This distinction will matter every time we want to take an expectation and set the stochastic integral to zero.

Lecture 12

Itô's Lemma for Brownian Motion

12.1 The missing fundamental theorem

At this point we understand well what is meant by

$$X_t = \int_0^t H_s dB_s : \quad (\star)$$

X_t is a continuous local martingale, enjoying properties formally similar to those of the Riemann integral.

For intuition about applications: in ordinary calculus, $df(t) = g(t) dt$ when $f(t) = \int_0^t g(s) ds$. In the same spirit we might write

$$dX_t = H_t dB_t$$

for $(\star)$. What we are still missing is a fundamental theorem of calculus, i.e. the analogue of

$$f(b(t)) - f(b(0)) = \int_0^t f'(b(s)) db(s).$$

This is what Itô's lemma supplies. In its simplest form, for $f \in C^2(\mathbb{R})$ (twice continuously differentiable),

$$f(B_t) - f(B_0) = \int_0^t \underbrace{f'(B_s)}_{\substack{\text{pathwise continuous,} \\ \text{adapted} \Rightarrow \text{predictable}}} dB_s + \frac{1}{2} \int_0^t f''(B_s) ds.$$

Example 12.1 (HW 6 #3). Take $f(x) = x^2$, so $f'(x) = 2x$ and $f''(B_s) = 2$. Then

$$\underbrace{B_t^2 - B_0^2}_{=B_t^2} = \int_0^t 2B_s dB_s + \frac{1}{2} \underbrace{\int_0^t 2 ds}_{=t},$$

so

$$B_t^2 - t = 2 \int_0^t B_s dB_s.$$

This is exactly the identity promised in Example 11.1.

12.2 The class $C^{1,2}$

Definition 12.2. We write $f \in C^{1,2}(\mathbb{R}^+ \times \mathbb{R})$ if

1. $f: (t, x) \mapsto f(t, x)$ is continuous;
2. $f_t = \partial_t f: (t, x) \mapsto f_t(t, x)$ exists and is continuous;
3. $f_x = \partial_x f: (t, x) \mapsto f_x(t, x)$ exists and is continuous;
4. $f_{xx} = \partial_{xx}^2 f: (t, x) \mapsto f_{xx}(t, x)$ exists and is continuous.

Note the asymmetry: one derivative in t , two in x . This is not an accident — it is forced by the fact that Brownian increments are of size $\sqrt{dt}$, so the second x -derivative contributes at the same order as the first t -derivative.

Theorem 12.3 (Itô's lemma for Brownian motion). *Let $f \in C^{1,2}(\mathbb{R}^+ \times \mathbb{R})$. Then*

$$f(t, B_t) - f(0, B_0) = \int_0^t f_x(s, B_s) dB_s + \int_0^t f_t(s, B_s) ds + \frac{1}{2} \int_0^t f_{xx}(s, B_s) ds.$$

Differential form

$$df(t, B_t) = f_x(t, B_t) dB_t + f_t(t, B_t) dt + \frac{1}{2} f_{xx}(t, B_t) dt.$$

This comes from the second-order Taylor expansion

$$\begin{aligned} f(t+dt, B_{t+dt}) - f(t, B_t) &\approx f_t(t, B_t) dt + f_x(t, B_t) dB_t \\ &\quad + \underbrace{\frac{1}{2} f_{tt}(t, B_t) (dt)^2 + f_{tx}(t, B_t) dt dB_t}_{\text{too small} = 0} \\ &\quad + \frac{1}{2} f_{xx}(t, B_t) \underbrace{(dB_t)^2}_{\Theta(dt)} + \text{h.o.t.} \end{aligned}$$

The key point is the last line: $(dB_t)^2$ is of order dt , not of order $(dt)^2$, so the second-order term in x survives while the second-order terms involving dt do not.

12.3 Proof of Theorem 12.3, part I

Fix t and let

$$t_k = k2^{-n}t, \quad k = 0, \dots, 2^n,$$

which quantizes $[0, t]$. Adding and subtracting the mixed term $f(t_{k-1}, B_{t_k})$,

$$f(t, B_t) - f(0, B_0) = \sum_{k=1}^{2^n} \left([f(t_k, B_{t_k}) - f(t_{k-1}, B_{t_k})] + [f(t_{k-1}, B_{t_k}) - f(t_{k-1}, B_{t_{k-1}})] \right). \quad (12.1)$$

We handle the two brackets separately: the first is a time increment at frozen space argument, the second a space increment at frozen time.

The time increment

By the mean value theorem there is $u_k \in (t_{k-1}, t_k)$ with

$$\begin{aligned} f(t_k, B_{t_k}) - f(t_{k-1}, B_{t_k}) &= f_t(u_k, B_{t_k})(t_k - t_{k-1}) \\ &= f_t(t_{k-1}, B_{t_k})(t_k - t_{k-1}) \\ &\quad + \underbrace{(f_t(u_k, B_{t_k}) - f_t(t_{k-1}, B_{t_k}))(t_k - t_{k-1})}_{\max_k \rightarrow 0 \text{ as } n \rightarrow \infty}. \end{aligned}$$

The error term vanishes because, on the event that the path stays in a compact set, $(r, s) \mapsto f_t(r, B_s)$ is uniformly continuous on $[0, t]^2$ and $|u_k - t_{k-1}| \leq t2^{-n} \rightarrow 0$. Summing the leading terms,

$$\sum_{k=1}^{2^n} f_t(t_{k-1}, B_{t_k})(t_k - t_{k-1}) \longrightarrow \int_0^t f_t(s, B_s) ds.$$

The space increment

By Taylor's theorem there is $v_k \in (t_{k-1}, t_k)$ with

$$\begin{aligned} f(t_{k-1}, B_{t_k}) - f(t_{k-1}, B_{t_{k-1}}) &= f_x(t_{k-1}, B_{t_{k-1}})(B_{t_k} - B_{t_{k-1}}) \\ &\quad + \frac{1}{2} f_{xx}(t_{k-1}, B_{t_{k-1}})(B_{t_k} - B_{t_{k-1}})^2 \\ &\quad + \frac{1}{2} (f_{xx}(t_{k-1}, B_{v_k}) - f_{xx}(t_{k-1}, B_{t_{k-1}}))(B_{t_k} - B_{t_{k-1}})^2. \end{aligned}$$

We continue with these three sums next time: the first converges to the Itô integral, the second to $\frac{1}{2} \int_0^t f_{xx}(s, B_s) ds$ via the quadratic variation of Brownian motion, and the third vanishes.

Lecture 13

Itô's Lemma and Itô Processes

13.1 Proof of Itô's lemma, part II

Recall the statement: for $f \in C^{1,2}$,

$$f(t, B_t) - f(0, B_0) = \int_0^t f_x(s, B_s) dB_s + \int_0^t f_t(s, B_s) ds + \frac{1}{2} \int_0^t f_{xx}(s, B_s) ds.$$

Proof of Theorem 12.3, continued. From last time, the decomposition (12.1) reads

$$\sum_{k=1}^{2^n} \left(\underbrace{[f(t_k, B_{t_k}) - f(t_{k-1}, B_{t_k})] + [f(t_{k-1}, B_{t_k}) - f(t_{k-1}, B_{t_{k-1}})]}_{\xrightarrow{n \rightarrow \infty} \int_0^t f_t(s, B_s) ds} \right),$$

and by Taylor's theorem the second bracket equals

$$\underbrace{f_x(t_{k-1}, B_{t_{k-1}}) \Delta_k B}_{=: A_k} + \frac{1}{2} f_{xx}(t_{k-1}, B_{t_{k-1}}) (\Delta_k B)^2 + \frac{1}{2} (f_{xx}(t_{k-1}, B_{v_k}) - f_{xx}(t_{k-1}, B_{t_{k-1}})) (\Delta_k B)^2$$

for some $v_k \in (t_{k-1}, t_k)$, where we abbreviate $\Delta_k B := B_{t_k} - B_{t_{k-1}}$ and $\Delta_k t := t_k - t_{k-1} = t 2^{-n}$.

Step 1: the A_k sum converges to the Itô integral

Localize so that the integrand is bounded. Define

$$\tau_m := \inf\{t \geq 0 : |B_t| \geq m\}$$

and set

$$H_s^{(m,n)} := \sum_{k=1}^{2^n} f_x(t_{k-1}, B_{t_{k-1}}) \mathbf{1}_{(t_{k-1} \wedge \tau_m, t_k \wedge \tau_m]}(s), \quad s \geq 0.$$

This is simple predictable, and by construction

$$\int H_s^{(m,n)} dB = \sum_{k=1}^{2^n} f_x(t_{k-1}, B_{t_{k-1}}) (B_{t_k \wedge \tau_m} - B_{t_{k-1} \wedge \tau_m}).$$

For every m , continuity of the paths and of f_x gives

$$H_s^{(m,n)} \xrightarrow{n \rightarrow \infty} \mathbf{1}_{[0, t \wedge \tau_m]}(s) f_x(s, B_s) \quad \text{a.s.}$$

By dominated convergence (the integrand is bounded on $\{|B| \leq m\}$) together with the Itô isometry,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^{2^n} f_x(t_{k-1}, B_{t_{k-1}}) (B_{t_k \wedge \tau_m} - B_{t_{k-1} \wedge \tau_m}) \stackrel{L^2}{=} \int_0^{t \wedge \tau_m} f_x(s, B_s) dB_s.$$

Letting $m \rightarrow \infty$ and passing to a subsequence $n = (n_j)_{j \geq 1}$,

$$\sum_{k=1}^{2^n} A_k \longrightarrow \int_0^t f_x(s, B_s) dB_s \quad \text{a.s.} \quad \checkmark$$

Step 2: the f_{xx} sum converges to the time integral

Split the second term by adding and subtracting $\Delta_k t$:

$$\sum_{k=1}^{2^n} \left[\underbrace{f_{xx}(t_{k-1}, B_{t_{k-1}}) \Delta_k t}_{\longrightarrow \int_0^t f_{xx}(s, B_s) ds} + f_{xx}(t_{k-1}, B_{t_{k-1}}) ((\Delta_k B)^2 - \Delta_k t) \right].$$

The first sum is a Riemann sum for a continuous integrand, so it converges pathwise. For the second, localize again: on $\{\tau_m \geq t\}$ we have $|f_{xx}(t_{k-1}, B_{t_{k-1}})| \leq K_m$ where $K_m := \sup\{|f_{xx}(r, x)| : r \in [0, t], |x| \leq m\}$. The increments $(\Delta_k B)^2 - \Delta_k t$ are independent and mean zero, so the cross terms drop and

$$\begin{aligned} \mathbb{E} \left(\left(\sum_{k=1}^{2^n} \mathbf{1}_{\{\tau_m \geq t\}} f_{xx}(t_{k-1}, B_{t_{k-1}}) ((\Delta_k B)^2 - \Delta_k t) \right)^2 \right) &\leq K_m^2 \mathbb{E} \left(\left(\sum_{k=1}^{2^n} ((\Delta_k B)^2 - \Delta_k t) \right)^2 \right) \\ &= K_m^2 \sum_{k=1}^{2^n} \mathbb{E} \left(((\Delta_k B)^2 - \Delta_k t)^2 \right) \\ &= 2t^2 K_m^2 2^{-n} \longrightarrow 0, \end{aligned}$$

using $\mathbb{E}(((\Delta_k B)^2 - \Delta_k t)^2) = 2(\Delta_k t)^2$ and $\Delta_k t = t 2^{-n}$. So this sum vanishes in probability as $n \rightarrow \infty$. $\checkmark$

Step 3: the remainder vanishes

Finally,

$$\begin{aligned} \left| \sum_{k=1}^{2^n} (f_{xx}(t_{k-1}, B_{v_k}) - f_{xx}(t_{k-1}, B_{t_{k-1}})) (\Delta_k B)^2 \right| &\leq \underbrace{\max_k |f_{xx}(t_{k-1}, B_{v_k}) - f_{xx}(t_{k-1}, B_{t_{k-1}})|}_{\longrightarrow 0 \text{ a.s.}} \\ &\quad \cdot \underbrace{\sum_{k=1}^{2^n} (\Delta_k B)^2}_{\longrightarrow t \text{ in probability}}. \end{aligned}$$

The first factor goes to 0 by uniform continuity of f_{xx} on compacts and continuity of the Brownian path; the second converges to the quadratic variation of Brownian motion on $[0, t]$. Their product therefore vanishes, completing the proof. $\square$

13.2 The “box calculus”

In differential form, Itô's lemma says

$$df(t, B_t) = f_x(t, B_t) dB_t + f_t(t, B_t) dt + \frac{1}{2} f_{xx}(t, B_t) dt.$$

On the other hand, for smooth f a formal Taylor expansion gives

$$\begin{aligned} df(t, B_t) &= f_x(t, B_t) dB_t + f_t(t, B_t) dt + \frac{1}{2} f_{xx}(t, B_t) (dB_t)^2 \\ &\quad + \frac{1}{2} f_{tt}(t, B_t) (dt)^2 + f_{tx}(t, B_t) dt dB_t + \text{h.o.t.} \end{aligned}$$

Matching the two expressions gives the multiplication rules

$$\boxed{(dB_t)^2 = dt, \quad dt dB_t = 0, \quad (dt)^2 = 0.}$$

This is a genuinely useful mnemonic: it turns Itô's lemma into ordinary Taylor expansion plus a multiplication table.

13.3 Itô processes

Definition 13.1.

$$\mathcal{L}_{\text{loc}}^1 := \left\{ H \text{ predictable} : \int_0^t |H_s| ds < \infty \text{ a.s. } \forall t \geq 0 \right\}.$$

Definition 13.2 (Itô process). An *Itô process* $(X_t)_{t \geq 0}$ is a process admitting the representation

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dB_s, \quad t \geq 0,$$

for $\mu \in \mathcal{L}_{\text{loc}}^1$, $\sigma \in \mathcal{L}_{\text{loc}}^2$, and X_0 an $\mathcal{F}_0$ -measurable random variable. We call μ the *drift* and σ the *diffusion*.

Remark 13.3. By Theorem 12.3, $f(t, B_t)$ is an Itô process for every $f \in C^{1,2}$: the drift is $f_t + \frac{1}{2} f_{xx}$ and the diffusion is f_x , both evaluated along the path.

Definition 13.4 (Integration against an Itô process). For an Itô process $(X_t)_{t \geq 0}$ with drift μ and diffusion σ , define

$$\int_0^t H_s dX_s := \int_0^t \mu_s H_s ds + \int_0^t \sigma_s H_s dB_s,$$

provided $\mu H \in \mathcal{L}_{\text{loc}}^1$ and $\sigma H \in \mathcal{L}_{\text{loc}}^2$.

Remark 13.5. This is a good definition for two reasons.

1. It is exactly what the differential notation demands: $dX_t = \mu_t dt + \sigma_t dB_t$ should give $H_t dX_t = H_t \mu_t dt + H_t \sigma_t dB_t$.
2. One can instead start from first principles: approximate $\int H dX$ for $H = \mathbf{1}_R$ with $R \in \mathcal{P}$, then extend by linearity and take limits, as we did for the Itô integral itself. The two constructions agree.

13.4 Itô's lemma for Itô processes

Theorem 13.6 (Itô's lemma, general form). *Let $(X_t)_{t \geq 0}$ be an Itô process with drift μ and diffusion σ , and let $f \in C^{1,2}$. Then*

$$f(t, X_t) - f(0, X_0) = \int_0^t f_x(s, X_s) dX_s + \int_0^t f_t(s, X_s) ds + \frac{1}{2} \int_0^t f_{xx}(s, X_s) \sigma_s^2 ds.$$

The only change from the Brownian case is the σ_s^2 in the last term — which is exactly what the box calculus predicts, since $(dX_t)^2 = (\mu_t dt + \sigma_t dB_t)^2 = \sigma_t^2 dt$.

Lecture 14

Applications of Itô's Lemma

14.1 Recap and proof

An Itô process $(X_t)_{t \geq 0}$ is defined by

$$X_t = \underbrace{X_0}_{\mathcal{F}_0\text{-mbl.}} + \int_0^t \underbrace{\mu_s}_{\text{drift}} ds + \int_0^t \underbrace{\sigma_s}_{\text{diffusion}} dB_s, \quad \text{i.e.} \quad dX_t = \mu_t dt + \sigma_t dB_t.$$

Proof of Theorem 13.6, for $f \in C^2$. We want

$$df(t, X_t) = f_x(t, X_t) dX_t + f_t(t, X_t) dt + \frac{1}{2} f_{xx}(t, X_t) \sigma_t^2 dt.$$

Taylor expanding,

$$\begin{aligned} df(t, X_t) &\approx f_t(t, X_t) dt + f_x(t, X_t) dX_t + \frac{1}{2} f_{tt}(t, X_t) (dt)^2 \xrightarrow{0} 0 \\ &\quad + f_{tx}(t, X_t) dt dX_t \xrightarrow{0} 0 + \frac{1}{2} f_{xx}(t, X_t) \underbrace{(dX_t)^2}_{= \sigma_t^2 dt} + \text{h.o.t.}, \end{aligned}$$

where the two cancellations come from the box calculus:

$$\begin{aligned} dt dX_t &= dt(\mu_t dt + \sigma_t dB_t) = 0, \\ dX_t dX_t &= (\mu_t dt + \sigma_t dB_t)^2 = \sigma_t^2 dt. \end{aligned}$$

□

14.2 The Black–Scholes asset model

Example 14.1 (Black–Scholes). Let $(X_t)_{t \geq 0}$ be a positive-valued Itô process with

$$dX_t = \underbrace{\mu}_{\text{interest}} X_t dt + \underbrace{\sigma}_{\text{volatility}} X_t dB_t, \quad \mu, \sigma \in \mathbb{R},$$

where X_t is interpreted as the price of an asset at time t .

Take $f(x) = \log(x)$, so $f'(x) = 1/x$ and $f''(x) = -1/x^2$. Itô's lemma gives

$$\begin{aligned} d\log(X_t) &= \frac{1}{X_t} dX_t - \frac{1}{2X_t^2} (\sigma X_t)^2 dt \\ &= \frac{1}{X_t} (\mu X_t dt + \sigma X_t dB_t) - \frac{1}{2} \sigma^2 dt \\ &= \sigma dB_t + \left(\mu - \frac{1}{2} \sigma^2 \right) dt. \end{aligned}$$

Integrating,

$$\begin{aligned} \log(X_t) &= \log(X_0) + \int_0^t \sigma dB_s + \int_0^t \left(\mu - \frac{1}{2} \sigma^2 \right) ds \\ &= \log(X_0) + \sigma B_t + \left(\mu - \frac{1}{2} \sigma^2 \right) t, \end{aligned}$$

so

$$X_t = X_0 e^{\mu t} \underbrace{e^{\sigma B_t - \frac{\sigma^2}{2} t}}_{\text{geometric BM}}.$$

The $-\frac{1}{2}\sigma^2 t$ in the exponent is pure Itô correction: naive calculus would have predicted $X_t = X_0 e^{\mu t + \sigma B_t}$, which has the wrong mean.

14.3 The Ornstein–Uhlenbeck process

Example 14.2 (Ornstein–Uhlenbeck). Consider

$$dX_t = -\theta X_t dt + \sigma dB_t, \quad \theta > 0, \sigma \in \mathbb{R}.$$

Take $f(t, x) = xe^{\theta t}$, so $f_t = \theta xe^{\theta t}$, $f_x = e^{\theta t}$, and $f_{xx} = 0$. Itô's lemma gives

$$\begin{aligned} X_t e^{\theta t} &= X_0 + \int_0^t e^{\theta s} dX_s + \int_0^t \theta X_s e^{\theta s} ds \\ &= X_0 + \int_0^t e^{\theta s} (-\theta X_s ds + \sigma dB_s) + \int_0^t \theta X_s e^{\theta s} ds \\ &= X_0 + \int_0^t e^{\theta s} \sigma dB_s, \end{aligned}$$

so

$$X_t = e^{-\theta t} X_0 + \sigma \int_0^t e^{-\theta(t-s)} dB_s.$$

Remark 14.3. Think of this as the measured voltage across a discharging capacitor, plus noise: the $e^{-\theta t}$ term is the deterministic decay of the initial condition, and the integral is an exponentially weighted average of the noise, with the most recent noise weighted most heavily.

14.4 Towards Fokker–Planck

Example 14.4 (Fokker–Planck equation). Let X_t satisfy the SDE

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t.$$

Q. What is the density of X_t ?

The distribution of X_t is characterized by $\mathbb{E}(f(X_t))$ for f smooth and compactly supported. Apply Itô's lemma to such an f (no explicit t dependence, so $f_t \equiv 0$) and take expectations:

$$\begin{aligned} \mathbb{E}(f(X_t)) - \mathbb{E}(f(X_0)) &= \mathbb{E} \left(\int_0^t f_x(X_s) \left(\mu(s, X_s) ds + \cancel{\sigma(s, X_s) dB_s} \right) \right. \\ &\quad \left. + \frac{1}{2} \int_0^t f_{xx}(X_s) \sigma(s, X_s)^2 ds \right) \\ &= \int_0^t \mathbb{E} \left(f_x(X_s) \mu(s, X_s) + \frac{1}{2} f_{xx}(X_s) \sigma(s, X_s)^2 \right) ds. \end{aligned}$$

The stochastic integral drops because f_x is bounded with compact support, so the integrand is in $\mathcal{L}^2$ and the integral is a genuine mean-zero martingale. Fubini moves the expectation inside. Differentiating in t ,

$$\frac{d}{dt} \mathbb{E}(f(X_t)) = \mathbb{E} \left(f_x(X_t) \mu(t, X_t) + \frac{1}{2} f_{xx}(X_t) \sigma(t, X_t)^2 \right).$$

We continue next time by converting this into a PDE for the density.

Lecture 15

Fokker–Planck and Langevin Dynamics

Throughout, $dX_t = \mu_t dt + \sigma_t dB_t$ and, for $f \in C^{1,2}$,

$$f(t, X_t) - f(0, X_0) = \int_0^t f_x(s, X_s) dX_s + \int_0^t f_t(s, X_s) ds + \frac{1}{2} \int_0^t f_{xx}(s, X_s) \sigma_s^2 ds.$$

15.1 The Fokker–Planck equation

Example 15.1 (Fokker–Planck, continued). Let $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t$ and suppose X_t has density p_t .

Q. How do we find the density of X_t ?

A. Apply Itô's lemma to a smooth compactly supported $f(x)$ and take expectations.

From last time,

$$\frac{d}{dt} \mathbb{E}(f(X_t)) = \mathbb{E}(f_x(X_t) \mu(t, X_t)) + \frac{1}{2} \mathbb{E}(f_{xx}(X_t) \sigma^2(t, X_t)).$$

Writing both sides against the density and integrating by parts (all boundary terms vanish since f has compact support),

$$\begin{aligned} \frac{d}{dt} \mathbb{E}(f(X_t)) &= \int p_t(x) f_x(x) \mu(t, x) dx + \frac{1}{2} \int p_t(x) f_{xx}(x) \sigma^2(t, x) dx \\ &= - \int [p_t(x) \mu(t, x)]' f(x) dx + \frac{1}{2} \int [p_t(x) \sigma^2(t, x)]'' f(x) dx. \end{aligned}$$

On the other hand the left-hand side is $\int \frac{d}{dt} p_t(x) f(x) dx$. Since f was an arbitrary smooth compactly supported test function, the two integrands must agree:

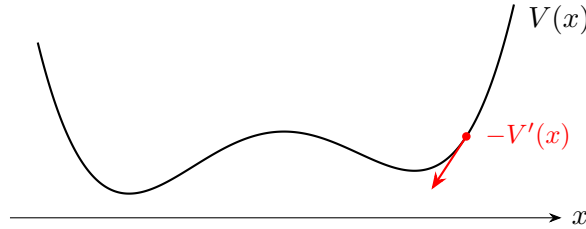
$$\partial_t p_t(x) = -\partial_x(p_t(x) \mu(t, x)) + \frac{1}{2} \partial_{xx}^2(p_t(x) \sigma^2(t, x)).$$

15.2 Langevin dynamics

Example 15.2 (Langevin dynamics / continuous-time gradient descent). This is the continuous-time limit of the unadjusted Langevin algorithm (ULA):

$$dX_t = -V'(X_t) dt + \sqrt{2/\beta} dB_t,$$

where $V \in C^2$ is bounded below with $V(x) \rightarrow \infty$ as $|x| \rightarrow \infty$. The drift pushes the particle downhill along the potential V ; the noise lets it escape local minima.



Here $\mu(t, x) = -V'(x)$ and $\sigma^2 = 2/\beta$, so the Fokker–Planck equation reads

$$\partial_t p_t = \partial_x (p_t V') + \frac{1}{\beta} \partial_{xx}^2 p_t.$$

If $p_0(x) \propto e^{-\beta V(x)}$, then $p_t(x) \propto e^{-\beta V(x)}$ for all t : the Gibbs measure solves the Fokker–Planck PDE stationarily. Indeed, for $p \propto e^{-\beta V}$ we have $\partial_x p = -\beta V' p$, so $\frac{1}{\beta} \partial_{xx}^2 p = -\partial_x (V' p)$ and the right-hand side vanishes.

Moreover, if either

1. V is convex; or
2. $e^{-\beta V}$ satisfies a Poincaré inequality, i.e. for $d\nu(x) \propto e^{-\beta V(x)} dx$,

$$\text{Var}_\nu(f) \leq C \mathbb{E}_\nu |\nabla f|^2 \quad \forall f,$$

then

$$X_t \xrightarrow{\text{law}} \nu \propto e^{-\beta V}.$$

Remark 15.3. This is why Langevin dynamics is a sampling algorithm: run the SDE long enough and the state is distributed according to the Gibbs measure $\propto e^{-\beta V}$. Taking $\beta \rightarrow \infty$ concentrates that measure on the minimizers of V , recovering gradient descent.

15.3 The multidimensional Itô formula

Let B_t be a d -dimensional standard Brownian motion, with $\sigma_t \in \mathbb{R}^{d \times d}$ and $\mu_t \in \mathbb{R}^d$.

Theorem 15.4 (Itô's lemma, multidimensional).

$$f(t, X_t) - f(0, X_0) = \int_0^t \nabla f(s, X_s) \cdot dX_s + \int_0^t f_t(s, X_s) ds + \frac{1}{2} \int_0^t \langle dX_s, \nabla^2 f(s, X_s) dX_s \rangle.$$

The last term is read through the box calculus: $dX_s dX_s^\top = \sigma_s \sigma_s^\top ds$, so it equals

$$\frac{1}{2} \int_0^t \text{tr}(\sigma_s^\top \nabla^2 f(s, X_s) \sigma_s) ds.$$

Example 15.5 (Itô product rule). Let X_t, Y_t be Itô processes. Then

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + dX_t dY_t.$$

Proof. Take $f(x, y) = xy$, so $\nabla f = (y, x)^\top$ and $\nabla^2 f = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Itô's lemma gives

$$d(X_t Y_t) = \left\langle \begin{bmatrix} Y_t \\ X_t \end{bmatrix}, \begin{bmatrix} dX_t \\ dY_t \end{bmatrix} \right\rangle + \frac{1}{2} \left\langle \begin{bmatrix} dX_t \\ dY_t \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} dX_t \\ dY_t \end{bmatrix} \right\rangle,$$

and the second term is $\frac{1}{2}(2 dX_t dY_t) = dX_t dY_t$. $\square$

Note the extra term relative to the ordinary product rule — it is exactly the cross-variation of X and Y .

15.4 Further examples

Example 15.6 (Brownian bridge). Let $(W_t)_{t \in [0,1]}$ be a Brownian bridge, so $W_0 = W_1 = 0$. One representation is

$$W_t = \underbrace{(1-t)}_{=: X_t} \underbrace{\int_0^t \frac{1}{1-s} dB_s}_{=: Y_t}, \quad t \in [0, 1).$$

Here $dX_t = -dt$ and $dY_t = \frac{1}{1-t} dB_t$, so $dX_t dY_t = 0$ and the product rule gives

$$dW_t = X_t dY_t + Y_t dX_t = (1-t) \cdot \frac{1}{1-t} dB_t - Y_t dt = -\frac{1}{1-t} W_t dt + dB_t,$$

using $Y_t = W_t/(1-t)$. The drift $-W_t/(1-t)$ is a restoring force that grows without bound as $t \uparrow 1$, pinning the path to 0 at the endpoint.

Example 15.7 (“Area” under a stopped Brownian motion). Let $\tau := \inf\{t \geq 0 : B_t \notin (-A, B)\}$ for $A, B > 0$.

Q. What is $\mathbb{E} \left(\int_0^\tau B_s ds \right)$?

Take $f(t, x) = tx$, so $f_t = x$, $f_x = t$, $f_{xx} = 0$. Itô's lemma gives

$$\tau B_\tau = \int_0^\tau s dB_s + \int_0^\tau B_s ds,$$

and the stochastic integral has mean zero (the integrand is bounded by τ on $[0, \tau]$ and $\mathbb{E}(\tau) = AB < \infty$), so

$$\mathbb{E} \left(\int_0^\tau B_s ds \right) = \mathbb{E}(\tau B_\tau) = \frac{1}{3} AB(B - A).$$

Remark 15.8 (Evaluating $\mathbb{E}(\tau B_\tau)$). The value $\frac{1}{3}AB(B-A)$ is cleanest to get from a second application of Itô's lemma, with $f(x) = x^3$:

$$B_t^3 = 3 \int_0^t B_s^2 dB_s + 3 \int_0^t B_s ds \implies \mathbb{E} \left(\int_0^\tau B_s ds \right) = \frac{1}{3} \mathbb{E}(B_\tau^3).$$

Since $B_\tau \in \{-A, B\}$ with $\mathbb{P}(B_\tau = B) = \frac{A}{A+B}$ and $\mathbb{P}(B_\tau = -A) = \frac{B}{A+B}$,

$$\mathbb{E}(B_\tau^3) = \frac{AB^3 - A^3B}{A+B} = \frac{AB(B^2 - A^2)}{A+B} = AB(B-A),$$

which gives the stated value.

Example 15.9 (Maximum of a Brownian motion with negative drift). Let

$$dX_t = -\mu dt + \sqrt{2} dB_t \implies X_t = -\mu t + \sqrt{2} B_t,$$

with $\mu > 0$. Then

$$\mathbb{P} \left\{ \sup_{t \geq 0} X_t \geq \lambda \right\} = e^{-\mu\lambda},$$

i.e. $\sup_{t \geq 0} X_t \sim \text{Exp}(\mu)$.

Lecture 16

Martingale Representation

16.1 The martingale representation theorem

Theorem 16.1 (MRT). *Let $(\mathcal{F}_t)_{t \geq 0}$ be the standard filtration generated by a standard Brownian motion $(B_t)_{t \geq 0}$. If $X \in L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ for some $T > 0$, then there exists $H \in \mathcal{L}^2$ such that*

$$X = \mathbb{E}(X) + \int_0^T H_s dB_s \quad a.s.$$

So every square-integrable random variable measurable with respect to the Brownian filtration is a stochastic integral against B , up to its mean. Brownian motion is, in this sense, the only source of randomness in its own filtration.

The proof rests on a density statement.

Lemma 16.2. *For every $T > 0$, the random variables of the form*

$$Z(\pi, \theta, n) = \prod_{j=1}^n e^{i\theta_j(B_{t_j} - B_{t_{j-1}})},$$

parametrized by $n \geq 1$, $\theta \in \mathbb{R}^n$, and partitions $\pi = \{t_0, \dots, t_n\}$ of $[0, T]$, have linear span dense in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$.

Proof of Theorem 16.1. From homework, for all $t \geq s \geq 0$,

$$e^{i\theta(B_t - B_s)} = e^{-\frac{1}{2}\theta^2(t-s)} + i\theta \int_s^t e^{-\frac{1}{2}\theta^2(t-u) + i\theta(B_u - B_s)} dB_u.$$

Applying this to each factor,

$$Z(\pi, \theta, n) = \prod_{j=1}^n \left(e^{-\frac{1}{2}\theta_j^2(t_j - t_{j-1})} + i\theta_j \int_{t_{j-1}}^{t_j} e^{-\frac{1}{2}\theta_j^2(t_j - u) + i\theta_j(B_u - B_{t_{j-1}})} dB_u \right),$$

and expanding the product with the Itô product rule (Example 15.5) gives

$$Z(\pi, \theta, n) = \mathbb{E}(Z(\pi, \theta, n)) + \int_0^T H_s dB_s \quad \text{for some } H \in \mathcal{L}^2.$$

So every generator has the desired representation, and by linearity so does every element of

$$\mathcal{S} := \text{span}\{Z(\pi, \theta, n) : \theta, \pi, n\}.$$

By Lemma 16.2, $\mathcal{S}$ is dense in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$, so there exist $(X_n)_{n \geq 1} \subset \mathcal{S}$ with $X_n \rightarrow X$ in L^2 , and hence

$$X_n - \mathbb{E}(X_n) \longrightarrow X - \mathbb{E}(X) \quad \text{in } L^2.$$

Writing $X_n = \mathbb{E}(X_n) + \int_0^T H_s^{(n)} dB_s$, convergence in L^2 of the left-hand sides means $(H^{(n)})_{n \geq 1}$ is Cauchy in $\mathcal{L}^2$ (by the Itô isometry), hence $H^{(n)} \rightarrow G$ in $\mathcal{L}^2$ for some G , and

$$X - \mathbb{E}(X) = \int_0^T G_s dB_s.$$

□

16.2 Application: replication of derivatives

Example 16.3 (Replication in a Black–Scholes market). Consider a financial market consisting of a risk-free bond,

$$dR_t = \mu R_t dt \implies R_t = R_0 e^{\mu t},$$

and a Black–Scholes asset,

$$dS_t = \mu S_t dt + \sigma S_t dB_t.$$

For every “derivative” $X \in L^2(\Omega, \mathcal{F}_T, \mathbb{P})$ there is a trading strategy (α_t, η_t) — where α_t is the number of shares of stock and η_t the number of bonds held at time t — that *replicates* X :

$$V_T = X \quad \text{a.s.}, \quad V_0 = e^{-\mu T} \mathbb{E}(X),$$

where

$$V_t = \alpha_t S_t + \eta_t R_t$$

is the portfolio value at time t . The strategy is required to be *self-financing*:

$$dV_t = \alpha_t dS_t + \eta_t dR_t,$$

i.e. the portfolio value changes only through price movements, with no money added or withdrawn.

Remark 16.4. Writing the same μ for the bond rate and the asset drift amounts to working under the risk-neutral measure, where the discounted asset price $e^{-\mu t} S_t$ is a martingale. That is what makes $V_0 = e^{-\mu T} \mathbb{E}(X)$ the arbitrage-free price.

16.3 Representation of local martingales

The MRT concerns a single random variable at a fixed horizon. The process-level statement is the following.

Theorem 16.5. Let $\mathbb{F} := (\mathcal{F}_t)_{t \geq 0}$ be the standard filtration generated by a standard Brownian motion $(B_t)_{t \geq 0}$, and let $(X_t)_{t \geq 0}$ be a continuous local martingale with respect to $\mathbb{F}$. Then there exists $H \in \mathcal{L}_{\text{loc}}^2$ such that

$$X_t = X_0 + \int_0^t H_s dB_s \quad \forall t \geq 0, \text{ a.s.}$$

Remark 16.6. Every local martingale with respect to $\mathbb{F}$ admits a continuous version, so the continuity hypothesis costs nothing.

Proof of Theorem 16.5, bounded case. Without loss of generality $X_0 = 0$. Assume first that $(X_t)_{t \geq 0}$ is bounded, so that it is an honest L^2 -martingale.

By the MRT applied at each integer horizon, for every $n \geq 1$ there is $H^{(n)} \in \mathcal{L}^2$ with

$$X_n = \int_0^n H_s^{(n)} dB_s \quad \text{a.s.}$$

Since X is a martingale and $X_{n+1} = \int_0^{n+1} H_s^{(n+1)} dB_s$,

$$X_n = \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) = \int_0^n H_s^{(n+1)} dB_s \quad \text{a.s., } \forall n \geq 1,$$

using that the Itô integral is a martingale. Comparing the two representations of X_n and applying the Itô isometry,

$$\mathbb{E} \left(\int_0^n (H_s^{(n)} - H_s^{(n+1)})^2 ds \right) = 0,$$

so the $H^{(n)}$ agree (up to null sets) on the intervals where they overlap. Define

$$H_s := \sum_{n \geq 1} \mathbf{1}_{(n-1, n]}(s) H_s^{(n)} \quad \implies \quad X_n = \int_0^n H_s dB_s \quad \text{a.s.}$$

Finally, for $t \leq n$,

$$X_t = \mathbb{E}(X_n \mid \mathcal{F}_t) = \mathbb{E} \left(\int_0^n H_s dB_s \mid \mathcal{F}_t \right) = \int_0^t H_s dB_s.$$

□

The general (unbounded, merely local) case is obtained by localizing this argument; we do that next time.

Lecture 17

Quadratic Variation

Throughout, $(\mathcal{F}_t)_{t \geq 0}$ is the standard Brownian filtration.

17.1 Recap

Theorem 17.1 (MRT). *If $X \in L^2(\Omega, \mathcal{F}_T, \mathbb{P})$, then there exists $H \in \mathcal{L}^2$ with*

$$X = \mathbb{E}(X) + \int_0^T H_s dB_s.$$

Proof sketch. It boils down to Itô's lemma plus density of the complex exponentials $Z(\pi, \theta, n)$ in $L^2(\Omega, \mathcal{F}_T, \mathbb{P})$. $\square$

Example 17.2. Replication of financial derivatives; also optimal control, filtering, and so on.

Corollary 17.3. *If $(X_t)_{t \geq 0}$ is a local martingale with respect to the Brownian filtration, then it admits a continuous version.*

17.2 Representation of local martingales, general case

Theorem 17.4. *If $(X_t)_{t \geq 0}$ is a local martingale, then there exists $H \in \mathcal{L}_{\text{loc}}^2$ such that*

$$X_t = X_0 + \int_0^t H_s dB_s \quad \text{a.s. } \forall t \geq 0.$$

Proof. By Corollary 17.3 we may assume $(X_t)_{t \geq 0}$ is a continuous local martingale, and without loss of generality $X_0 = 0$. Let

$$\tau_k := \inf\{t \geq 0 : |X_t| = k\}, \quad k \geq 0,$$

with $\tau_0 := 0$. Then $|X_{t \wedge \tau_k}| \leq k$ and $\tau_k \rightarrow \infty$ a.s.

The stopped process X^{τ_k} is a bounded martingale, so by the bounded case proved last time there is $H^{(k)} \in \mathcal{L}^2$ with

$$X_{t \wedge \tau_k} = \int_0^t H_s^{(k)} dB_s = \int \mathbf{1}_{[0, t \wedge \tau_k]} H^{(k)} dB \quad \text{a.s. } \forall t \geq 0.$$

Applying the same identity at level $k + 1$ and stopping at $\tau_k \leq \tau_{k+1}$,

$$X_{t \wedge \tau_k} = X_{t \wedge \tau_k \wedge \tau_{k+1}} = \int_0^{t \wedge \tau_k} H_s^{(k+1)} dB_s = \int \mathbf{1}_{[0, t \wedge \tau_k]} H^{(k+1)} dB.$$

The two representations of $X_{t \wedge \tau_k}$ must agree, so by monotone convergence and the Itô isometry,

$$\mathbb{E} \left(\int_0^\infty \mathbf{1}_{[0, \tau_k]} (H_s^{(k+1)} - H_s^{(k)})^2 ds \right) \stackrel{\text{MCT}}{=} \lim_{t \rightarrow \infty} \mathbb{E} \left(\int_0^\infty \mathbf{1}_{[0, t \wedge \tau_k]} (H_s^{(k+1)} - H_s^{(k)})^2 ds \right) \stackrel{\text{Itô isom.}}{=} 0.$$

That is, consecutive integrands agree on $[0, \tau_k]$ up to a null set, so we may splice them together:

$$H_s := \sum_{k \geq 1} \mathbf{1}_{(\tau_{k-1}, \tau_k]}(s) H_s^{(k)} \in \mathcal{L}_{\text{loc}}^2.$$

Then $X_{t \wedge \tau_k} = \int \mathbf{1}_{[0, t \wedge \tau_k]} H dB$ for every k , and $(\tau_k)_{k \geq 1}$ is a localizing sequence for H , so by the definition of the Itô integral on $\mathcal{L}_{\text{loc}}^2$ (Theorem 11.6),

$$X_t = \int_0^t H_s dB_s \quad \text{a.s.}$$

□

17.3 Quadratic variation

Example 17.5 (Recall, from homework). Let $h: \mathbb{R}^+ \rightarrow \mathbb{R}$ be deterministic with $\int_{\mathbb{R}^+} h^2(s) ds = \infty$. Then

$$W_t = \int_0^{\tau_t} h(s) dB_s \quad \text{is a standard Brownian motion,}$$

where

$$\tau_t = \inf \left\{ u : \int_0^u h^2(s) ds > t \right\}, \quad t \geq 0.$$

So a stochastic integral can be turned back into a Brownian motion by a *deterministic time change*, and the clock that does it is $u \mapsto \int_0^u h^2$. Quadratic variation is what generalizes that clock.

Definition 17.6 (Quadratic variation). For a process $(X_t)_{t \geq 0}$ and a partition $\pi = \{t_0 = 0, \dots, t_k = t\}$ of $[0, t]$, define

$$Q_\pi(X) := \sum_{i=1}^k (X_{t_i} - X_{t_{i-1}})^2.$$

The *quadratic variation* of X , denoted $[X]$, is the process defined by

$$[X]_t = \lim_{\|\pi\| \rightarrow 0} Q_\pi(X),$$

where the limit (if it exists) is with respect to any sequence of partitions of $[0, t]$ with vanishing mesh, and is taken in probability.

Theorem 17.7. *Let X_t be an Itô process, $dX_t = \mu_t dt + \sigma_t dB_t$. Then the quadratic variation of X exists, admits a continuous version, and*

$$[X]_t = X_t^2 - X_0^2 - 2 \int_0^t X_s dX_s = \int_0^t \sigma_s^2 ds.$$

Remark 17.8. The first equality is Itô's lemma applied to $f(x) = x^2$, rearranged: the quadratic variation is precisely the correction term that distinguishes stochastic calculus from ordinary calculus. The second says the accumulated variation is driven entirely by the diffusion coefficient — the drift is too smooth to contribute.

Example 17.9. For Brownian motion, $\mu \equiv 0$ and $\sigma \equiv 1$, so

$$2 \int_0^t B_s dB_s = B_t^2 - \underbrace{B_0^2}_{=0} - t, \quad [B]_t = t \quad \text{a.s.}$$

Lecture 18

Quadratic Variation and Time Change

18.1 Quadratic variation, again

Definition 18.1 (Quadratic variation). For a partition $\pi = \{0 = t_0 < \cdots < t_k = t\}$ of $[0, t]$ put

$$Q_\pi(X) := \sum_{i=1}^k (X_{t_i} - X_{t_{i-1}})^2.$$

The *quadratic variation* of X , when it exists, is

$$[X]_t := \lim_{\|\pi\| \rightarrow 0} Q_\pi(X),$$

the limit being in probability along any sequence of partitions with vanishing mesh.

Pathwise, $[X]_t(\omega)$ measures a kind of variation present in the trajectory $t \mapsto X_t(\omega)$ — one that is invisible to ordinary calculus, since a differentiable path has zero quadratic variation.

Theorem 18.2. *Let $(X_t)_{t \geq 0}$ be an Itô process, $dX_t = \mu_t dt + \sigma_t dB_t$. Then the quadratic variation of X exists, admits a continuous version, and*

$$[X]_t = X_t^2 - X_0^2 - 2 \int_0^t X_s dX_s = \int_0^t \sigma_s^2 ds, \quad t \geq 0.$$

Example 18.3. For Brownian motion, $[B]_t = t$, equivalently

$$B_t^2 - t = 2 \int_0^t B_s dB_s.$$

Proof of Theorem 18.2. Step 1: assume X is bounded. Let π be a partition of $[0, t]$. Expanding the square and telescoping,

$$Q_\pi(X) = X_t^2 - X_0^2 - 2 \sum_{i=1}^k X_{t_{i-1}} (X_{t_i} - X_{t_{i-1}}),$$

which is the identity $(b - a)^2 = b^2 - a^2 - 2a(b - a)$ applied on each subinterval. Now substitute the Itô representation of each increment,

$$X_{t_i} - X_{t_{i-1}} = \int_{t_{i-1}}^{t_i} (\mu_s ds + \sigma_s dB_s),$$

and pull the $\mathcal{F}_{t_{i-1}}$ -measurable factor $X_{t_{i-1}}$ inside:

$$Q_\pi(X) = X_t^2 - X_0^2 - 2 \int_0^t X_s^\pi (\mu_s ds + \sigma_s dB_s), \quad (\star)$$

where

$$X_s^\pi := \sum_{i=1}^k X_{t_{i-1}} \mathbf{1}_{(t_{i-1}, t_i]}(s)$$

is the simple predictable approximation of X along π .

As $\|\pi\| \rightarrow 0$ we have $X_s^\pi \rightarrow X_s$ a.s., by continuity of X . Hence:

- by dominated convergence, $\lim_{\|\pi\| \rightarrow 0} \int_0^t X_s^\pi \mu_s ds = \int_0^t X_s \mu_s ds$;
- for the second integrand, $\int_0^t X_s^\pi \sigma_s dB_s \xrightarrow{\mathbb{P}} \int_0^t X_s \sigma_s dB_s$, by the Itô isometry together with dominated convergence (here boundedness of X is used).

So the right-hand side of $(\star)$ converges in probability, giving

$$Q_\pi(X) \xrightarrow{\mathbb{P}} X_t^2 - X_0^2 - 2 \int_0^t X_s dX_s.$$

The second equality of the theorem follows from Itô's lemma applied to $f(x) = x^2$:

$$X_t^2 = X_0^2 + 2 \int_0^t X_s dX_s + \frac{1}{2} \int_0^t 2\sigma_s^2 ds \implies X_t^2 - X_0^2 - 2 \int_0^t X_s dX_s = \int_0^t \sigma_s^2 ds.$$

In particular the limit is $\int_0^t \sigma_s^2 ds$, which is continuous in t .

Step 2: localize. Assume without loss of generality $X_0 = 0$, and put

$$\tau_m := \inf\{t \geq 0 : |X_t| = m\}, \quad X^{(m)} := (X_{t \wedge \tau_m})_{t \geq 0},$$

which is a bounded Itô process. On $\{t \leq \tau_m\}$ the two processes agree, so Step 1 applies to $X^{(m)}$ and gives

$$\mathbf{1}_{\{t \leq \tau_m\}} Q_\pi(X) = \mathbf{1}_{\{t \leq \tau_m\}} Q_\pi(X^{(m)}) \xrightarrow[\mathbb{P}]{\|\pi\| \rightarrow 0} \mathbf{1}_{\{t \leq \tau_m\}} \int_0^t \sigma_s^2 ds.$$

Since X has continuous paths, $\tau_m \rightarrow \infty$ a.s., so $\mathbb{P}(t \leq \tau_m) \rightarrow 1$ as $m \rightarrow \infty$ and therefore

$$Q_\pi(X) \xrightarrow{\mathbb{P}} \int_0^t \sigma_s^2 ds \quad \text{as } \|\pi\| \rightarrow 0. \quad \square$$

Remark 18.4. Read the two equalities in the theorem in opposite directions. The first is a *definition-level* statement: quadratic variation is exactly the discrepancy between Itô calculus and ordinary calculus. The second is a *computational* one: that discrepancy is completely determined by the diffusion coefficient, and the drift contributes nothing.

18.2 Random time change of local martingales

The formula $[X]_t = \int_0^t \sigma_s^2 ds$ suggests reading $[X]$ as a clock: it runs fast where the process is volatile and slowly where it is calm. The rest of this lecture builds the machinery to run a process on that clock.

Definition 18.5 (The σ -algebra at a stopping time). Let τ be a stopping time. Then

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : \{\tau \leq t\} \cap A \in \mathcal{F}_t \ \forall t \geq 0\} = \text{the information revealed up to time } \tau.$$

Proposition 18.6 (Optional sampling). Let $(X_t)_{t \geq 0}$ be a bounded continuous martingale and let $\nu \leq \tau$ be finite stopping times. Then

$$X_\nu = \mathbb{E}(X_\tau \mid \mathcal{F}_\nu).$$

Proof. First, X_τ is $\mathcal{F}_\tau$ -measurable: for every $\alpha \in \mathbb{R}$ and $t \geq 0$,

$$\{X_\tau \leq \alpha\} \cap \{\tau \leq t\} = \{X_{t \wedge \tau} \leq \alpha\} \cap \{\tau \leq t\} \in \mathcal{F}_t,$$

because $X_\tau = X_{t \wedge \tau}$ on $\{\tau \leq t\}$ and the stopped process is adapted. Hence $\{X_\tau \leq \alpha\} \in \mathcal{F}_\tau$.

Since X is a bounded martingale, the martingale convergence theorem provides X_∞ with $X_t \rightarrow X_\infty$ a.s. and in L^1 ; in particular $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t)$.

Now let $A \in \mathcal{F}_\tau$ and fix $t \geq 0$. Then $\mathbf{1}_A \mathbf{1}_{\{\tau \leq t\}}$ is $\mathcal{F}_{t \wedge \tau}$ -measurable, so

$$\begin{aligned} \mathbb{E}(X_\infty \mathbf{1}_{\{\tau \leq t\}} \mathbf{1}_A) &= \mathbb{E}\left(\mathbb{E}(X_\infty \mid \mathcal{F}_{t \wedge \tau}) \mathbf{1}_{\{\tau \leq t\}} \mathbf{1}_A\right) \\ &= \mathbb{E}(X_{t \wedge \tau} \mathbf{1}_{\{\tau \leq t\}} \mathbf{1}_A) \\ &= \mathbb{E}(X_\tau \mathbf{1}_{\{\tau \leq t\}} \mathbf{1}_A). \end{aligned}$$

The second equality is optional stopping at the bounded stopping time $t \wedge \tau$; the third is $X_{t \wedge \tau} = X_\tau$ on $\{\tau \leq t\}$. Let $t \rightarrow \infty$. As τ is finite, $\mathbf{1}_{\{\tau \leq t\}} \uparrow 1$, and X is bounded, so dominated convergence gives

$$\mathbb{E}(X_\infty \mathbf{1}_A) = \mathbb{E}(X_\tau \mathbf{1}_A) \quad \forall A \in \mathcal{F}_\tau,$$

that is, $X_\tau = \mathbb{E}(X_\infty \mid \mathcal{F}_\tau)$. The same argument applies to ν , and $\nu \leq \tau$ gives $\mathcal{F}_\nu \subseteq \mathcal{F}_\tau$, so by the tower property

$$X_\nu = \mathbb{E}(X_\infty \mid \mathcal{F}_\nu) = \mathbb{E}\left(\underbrace{\mathbb{E}(X_\infty \mid \mathcal{F}_\tau)}_{= X_\tau} \mid \mathcal{F}_\nu\right) = \mathbb{E}(X_\tau \mid \mathcal{F}_\nu). \quad \square$$

Theorem 18.7. Let $(X_t)_{t \geq 0}$ be a continuous local martingale and let $(\tau_t)_{t \geq 0}$ be an increasing family of finite stopping times. Then $(X_{\tau_t})_{t \geq 0}$ is a local martingale with respect to the filtration $(\mathcal{F}_{\tau_t})_{t \geq 0}$.

Proof. Put

$$\nu_n := \inf\{t \geq 0 : |X_t| = n\} \wedge n, \quad n \geq 1,$$

so that $(X_{t \wedge \nu_n})_{t \geq 0}$ is a continuous bounded martingale and $\nu_n \rightarrow \infty$ a.s. Applying Proposition 18.6 to it at the finite stopping times $\tau_s \leq \tau_t$,

$$X_{\tau_s \wedge \nu_n} = \mathbb{E}(X_{\tau_t \wedge \nu_n} \mid \mathcal{F}_{\tau_s}) \quad \forall 0 \leq s \leq t.$$

So each stopped process $(X_{t \wedge \nu_n})_{t \geq 0}$ is a martingale with respect to $(\mathcal{F}_{\tau_t})_{t \geq 0}$, and $(X_{\tau_t})_{t \geq 0}$ is a local martingale with localizing sequence $(\nu_n)_{n \geq 1}$. $\square$

18.3 Dubins–Schwarz

Theorem 18.8 (Dubins–Schwarz). *Let $X_t = \int_0^t \sigma_s dB_s$ for $\sigma \in \mathcal{L}_{\text{loc}}^2$ with*

$$[X]_\infty = \int_0^\infty \sigma_s^2 ds = \infty \quad \text{a.s.},$$

and set $\tau_t := \inf\{u \geq 0 : [X]_u > t\}$. Then $(X_{\tau_t})_{t \geq 0}$ is a standard Brownian motion with respect to the standard filtration $(\mathcal{F}_{\tau_t})_{t \geq 0}$.

Restated: there exists a standard Brownian motion $(W_t)_{t \geq 0}$ such that

$$X_t = W_{[X]_t} \quad \forall t \geq 0.$$

Remark 18.9. The hypothesis is on $[X]_\infty$, not on $[X]_t$ for finite t : we need the clock to run forever so that the time-changed process is defined on all of $[0, \infty)$.

Every continuous stochastic integral is therefore a Brownian motion *watched on the wrong clock*, and the clock is its own quadratic variation. We prove this next lecture.

Lecture 19

Change of Time

19.1 Recap

Theorem 19.1. *If $(X_t)_{t \geq 0}$ is a continuous local martingale and $(\tau_t)_{t \geq 0}$ is an increasing family of finite stopping times, then $(X_{\tau_t})_{t \geq 0}$ is a local martingale with respect to $(\mathcal{F}_{\tau_t})_{t \geq 0}$.*

Theorem 19.2 (Dubins–Schwarz). *Let $X_t = \int_0^t \sigma_s dB_s$ satisfy $[X]_\infty = \int_0^\infty \sigma_s^2 ds = \infty$ a.s. Then, with $\tau_t := \inf\{u \geq 0 : [X]_u > t\}$, the process $(X_{\tau_t})_{t \geq 0}$ is a standard Brownian motion with respect to the standard filtration $(\mathcal{F}_{\tau_t})_{t \geq 0}$.*

Example 19.3. Restated: there exists a standard Brownian motion W such that

$$X_t = W_{[X]_t}.$$

Every result about Brownian motion is therefore a result about continuous stochastic integrals, read on the clock $[X]$. Here is the prototype.

Example 19.4 (Law of the iterated logarithm).

$$\limsup_{t \rightarrow \infty} \frac{X_t}{\sqrt{2[X]_t \log \log [X]_t}} = 1 \quad \text{a.s.}$$

Indeed, substituting $X_t = W_{[X]_t}$ the left-hand side is

$$\limsup_{t \rightarrow \infty} \frac{W_{[X]_t}}{\sqrt{2[X]_t \log \log [X]_t}} = \limsup_{s \rightarrow \infty} \frac{W_s}{\sqrt{2s \log \log s}} = 1,$$

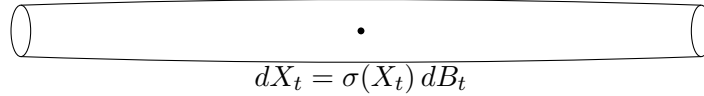
using $[X]_t \rightarrow \infty$ to reindex by $s = [X]_t$, and then the law of the iterated logarithm for Brownian motion.

19.2 Lévy's characterization

Theorem 19.5 (Lévy's characterization of Brownian motion). *Let $(X_t)_{t \geq 0}$ be a continuous local martingale with $X_0 = 0$ and quadratic variation $[X]_t = t$. Then $(X_t)_{t \geq 0}$ is a standard Brownian motion.*

This is the special case of Dubins–Schwarz in which the clock is already correct: if $X_t = \int_0^t \sigma_s dB_s$ has $[X]_t = t$, then $\tau_t = t$, and Theorem 19.2 says directly that X is a standard Brownian motion with respect to $(\mathcal{F}_t)_{t \geq 0}$.

Example 19.6 (The infinitely long tube). Picture a particle diffusing along an infinitely long tube whose temperature varies from place to place:



Here $\sigma^2(x)$ is proportional to the temperature at location x : hotter means more volatility, so the particle jitters faster in the hot regions and slower in the cold ones.

Dubins–Schwarz says we may speed up and slow down the clock so that every collision looks like it happens at the same rate — that is, so that the medium becomes homogeneous. On that clock the dynamics are simply

$$dX_t = \sigma dB_t.$$

The inhomogeneity of the tube is not a property of the particle’s path; it is a property of the clock we were reading it on.

19.3 Proof of Dubins–Schwarz

Proof of Theorem 19.2. Fix $\theta \in \mathbb{R}$ and define the (complex-valued) process

$$Z_t := \exp(i\theta X_t + \tfrac{1}{2}\theta^2 [X]_t) = \exp(i\theta X_t) \exp(\tfrac{1}{2}\theta^2 [X]_t).$$

By the Itô product rule,

$$dZ_t = e^{i\theta X_t} d\left(e^{\frac{1}{2}\theta^2 [X]_t}\right) + e^{\frac{1}{2}\theta^2 [X]_t} d\left(e^{i\theta X_t}\right) + d\left(e^{i\theta X_t}\right) d\left(e^{\frac{1}{2}\theta^2 [X]_t}\right).$$

The last term vanishes because $[X]$ has finite variation. For the other two, $[X]$ is again of finite variation, so $d\left(e^{\frac{1}{2}\theta^2 [X]_t}\right) = \frac{1}{2}\theta^2 e^{\frac{1}{2}\theta^2 [X]_t} d[X]_t$, while Itô’s lemma gives

$$d\left(e^{i\theta X_t}\right) = i\theta e^{i\theta X_t} dX_t - \tfrac{1}{2}\theta^2 e^{i\theta X_t} d[X]_t.$$

The two $d[X]_t$ terms cancel exactly, leaving

$$dZ_t = i\theta Z_t dX_t = i\theta Z_t \sigma_t dB_t.$$

So $(Z_t)_{t \geq 0}$ is a continuous local martingale, and by Theorem 19.1, $(Z_{\tau_t})_{t \geq 0}$ is a local martingale with respect to $(\mathcal{F}_{\tau_t})_{t \geq 0}$.

By construction of τ we have $[X]_{\tau_t} = t$, hence

$$Z_{\tau_t} = \exp(i\theta X_{\tau_t} + \tfrac{1}{2}\theta^2 t), \quad |Z_{\tau_t}| = e^{\frac{1}{2}\theta^2 t},$$

which is bounded on compact time intervals. A bounded local martingale is an honest martingale, so for $0 \leq s \leq t$,

$$\mathbb{E}\left(\exp(i\theta X_{\tau_t} + \tfrac{1}{2}\theta^2 t) \mid \mathcal{F}_{\tau_s}\right) = Z_{\tau_s} = \underbrace{\exp(i\theta X_{\tau_s} + \tfrac{1}{2}\theta^2 s)}_{\mathcal{F}_{\tau_s}\text{-measurable}}.$$

Rearranging,

$$\mathbb{E}\left(\exp(i\theta(X_{\tau_t} - X_{\tau_s})) \mid \mathcal{F}_{\tau_s}\right) = \exp\left(-\frac{1}{2}\theta^2(t - s)\right).$$

The right-hand side is deterministic and is the characteristic function of $\mathcal{N}(0, t - s)$; since θ was arbitrary,

$$X_{\tau_t} - X_{\tau_s} \sim \mathcal{N}(0, t - s) \quad \text{and} \quad X_{\tau_t} - X_{\tau_s} \perp\!\!\!\perp \mathcal{F}_{\tau_s}.$$

So $(X_{\tau_t})_{t \geq 0}$ is a Gaussian process with $X_{\tau_0} = 0$ and independent stationary increments. Two things remain to check: that $t \mapsto X_{\tau_t}$ is a.s. continuous, and that $(\mathcal{F}_{\tau_t})_{t \geq 0}$ is a standard filtration. The latter is Lemma 19.8 below. $\square$

Remark 19.7 (Continuity of the time-changed path). The only thing that could go wrong is a stretch on which the clock $[X]$ is flat, since there τ jumps. But on such a stretch $\int \sigma^2 ds$ does not move, so X itself is constant there — the jump in τ therefore costs nothing, and $t \mapsto X_{\tau_t}$ is well defined and continuous. The full argument is in the course notes.

19.4 The time-changed filtration is standard

Lemma 19.8. $(\mathcal{F}_{\tau_t})_{t \geq 0}$ is a standard filtration.

Proof. Completeness. $\mathcal{F}_{\tau_0} \supseteq \mathcal{F}_0$, which contains all $\mathbb{P}$ -null sets because $\mathbb{F}$ is standard. So $(\mathcal{F}_{\tau_t})_{t \geq 0}$ is complete.

Right continuity. The map $t \mapsto \tau_t$ is right continuous and nondecreasing, so for every $\alpha \geq 0$ and $n \geq 1$,

$$\{\tau_t < \alpha + \frac{1}{n}\} = \bigcup_{m \geq 1} \left\{ \tau_{t + \frac{1}{m}} < \alpha + \frac{1}{n} \right\}.$$

Let $A \in \mathcal{F}_{\tau_s}$ for every $s > t$. Then $A \in \mathcal{F}_{\tau_{t+1/m}}$ for every m , so

$$A \cap \{\tau_t < \alpha + \frac{1}{n}\} = \bigcup_{m \geq 1} \underbrace{\left(A \cap \left\{ \tau_{t + \frac{1}{m}} < \alpha + \frac{1}{n} \right\} \right)}_{\in \mathcal{F}_{\alpha + 1/n}} \in \mathcal{F}_{\alpha + \frac{1}{n}}.$$

Intersecting over n and using right continuity of $\mathbb{F}$,

$$A \cap \{\tau_t \leq \alpha\} = A \cap \bigcap_{n \geq 1} \{\tau_t < \alpha + \frac{1}{n}\} \in \bigcap_{n \geq 1} \mathcal{F}_{\alpha + \frac{1}{n}} = \mathcal{F}_{\alpha^+} = \mathcal{F}_\alpha.$$

As α was arbitrary, $A \in \mathcal{F}_{\tau_t}$. $\square$

Lecture 20

Change of Measure and Girsanov's Theorem

The theme of this lecture: rather than analyzing a hard process under the given measure, tilt the measure until the process becomes easy.

20.1 Absolute continuity

Definition 20.1. Let $\mathbb{P}, \mathbb{Q}$ be probability measures on $(\Omega, \mathcal{F})$. We write $\mathbb{Q} \ll \mathbb{P}$ ($\mathbb{Q}$ is *absolutely continuous* with respect to $\mathbb{P}$) if

$$\mathbb{P}(A) = 0 \implies \mathbb{Q}(A) = 0 \quad \forall A \in \mathcal{F}.$$

Theorem 20.2 (Radon–Nikodym). *If $\mathbb{Q} \ll \mathbb{P}$, then there exists $\Lambda \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ such that*

$$\mathbb{E}_{\mathbb{Q}}(\mathbf{1}_A) = \mathbb{E}_{\mathbb{P}}(\Lambda \mathbf{1}_A) \quad \forall A \in \mathcal{F}.$$

Changing measure from $\mathbb{P}$ to $\mathbb{Q}$ is exactly a matter of reweighting by Λ ; the standard notation is $\frac{d\mathbb{Q}}{d\mathbb{P}} = \Lambda$, so that

$$\mathbb{E}_{\mathbb{Q}}(\mathbf{1}_A) = \int_A d\mathbb{Q} = \int_A \frac{d\mathbb{Q}}{d\mathbb{P}} d\mathbb{P} = \mathbb{E}_{\mathbb{P}}\left(\frac{d\mathbb{Q}}{d\mathbb{P}} \mathbf{1}_A\right).$$

Example 20.3 (Neyman–Pearson). Consider testing

$$H_0 : (\Omega, \mathcal{F}, \mathbb{P}) \quad \text{against} \quad H_1 : (\Omega, \mathcal{F}, \mathbb{Q}).$$

The most powerful test is the likelihood ratio test,

$$\hat{H}(\omega) = \begin{cases} H_1 & \text{if } \Lambda(\omega) \geq \eta, \\ H_0 & \text{otherwise,} \end{cases} \quad \eta \geq 0.$$

So the Radon–Nikodym derivative *is* the statistic on which optimal detection is based.

Example 20.4 (Gaussian shift). Let $\mathbb{P} = \mathcal{N}(0, I)$ and $\mathbb{Q} = \mathcal{N}(m, I)$ on $\mathbb{R}^n$, with $m \in \mathbb{R}^n$. Comparing the two densities,

$$\Lambda(x) = \frac{d\mathbb{Q}}{d\mathbb{P}}(x) = \exp\left(-\frac{1}{2}|x - m|^2 + \frac{1}{2}|x|^2\right) = \exp\left(x^\top m - \frac{1}{2}|m|^2\right).$$

The last example is the whole of Girsanov's theorem in one line: shifting the mean of a Gaussian costs an exponential tilt that is *linear in the observation* minus half the energy of the shift. Girsanov is the infinite-dimensional version, with $x^\top m$ becoming $\int \mu dB$ and $|m|^2$ becoming $\int \mu^2 ds$.

20.2 Girsanov's theorem

Theorem 20.5 (Girsanov). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry a standard filtration $\mathbb{F}$ and an $\mathbb{F}$ -standard Brownian motion $(B_t)_{t \geq 0}$. Let $\mu \in \mathcal{L}_{\text{loc}}^2$ and set*

$$M_t := \exp \left(\int_0^t \mu_s dB_s - \frac{1}{2} \int_0^t \mu_s^2 ds \right), \quad t \geq 0.$$

Suppose $(M_t)_{t \geq 0}$ is a martingale on the time interval $[0, T]$ — equivalently, that $\mathbb{E}(M_T) = 1$. Then

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = M_T$$

defines a probability measure $\mathbb{Q} \ll \mathbb{P}$ on $(\Omega, \mathcal{F})$, and

$$X_t = B_t - \int_0^t \mu_s ds$$

is an $\mathbb{F}$ -standard Brownian motion under $\mathbb{Q}$ on $[0, T]$.

In words: an Itô process whose diffusion coefficient is 1 can have its drift removed entirely, at the cost of reweighting the underlying measure by M_T .

20.3 Applications

Risk-neutral pricing

Example 20.6. On $(\Omega, \mathcal{F}, \mathbb{P})$, consider a market with a risky asset and a risk-free bond,

$$dS_t = \mu_t S_t dt + \sigma S_t dB_t \quad (\mu \text{ predictable}), \quad dR_t = r R_t dt \quad (r > 0 \text{ deterministic}).$$

If $\mu_t \equiv r$, then any claim $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ has fair price $e^{-rT} \mathbb{E}_{\mathbb{P}}(X)$. In general $\mu_t \neq r$, so rewrite the asset dynamics to look as though it were:

$$dS_t = r S_t dt + \sigma S_t \underbrace{\left(dB_t - \frac{r - \mu_t}{\sigma} dt \right)}_{=: d\tilde{B}_t} = r S_t dt + \sigma S_t d\tilde{B}_t.$$

By Girsanov applied with drift $\frac{r - \mu_t}{\sigma}$, the process $(\tilde{B}_t)_{t \geq 0}$ is a standard Brownian motion under the measure $\mathbb{Q}$ defined by

$$d\mathbb{Q} = \exp \left(\int_0^T \frac{r - \mu_s}{\sigma} dB_s - \frac{1}{2} \int_0^T \left(\frac{r - \mu_s}{\sigma} \right)^2 ds \right) d\mathbb{P}.$$

Under $\mathbb{Q}$ we are back in the constant-drift setting, so the asset price is $e^{-rT} \mathbb{E}_{\mathbb{Q}}(X)$.

Hypothesis testing

Example 20.7 (Signal detection). On the basis of observations $(Y_t)_{t \geq 0}$, test

$$H_0 : dY_t = dB_t \quad \text{against} \quad H_1 : dY_t = \mu_t dt + dB_t,$$

where μ_t is known and deterministic with $\int_0^T \mu_t^2 dt < \infty$. Girsanov identifies the likelihood ratio explicitly:

$$\frac{d\mathbb{P}_1}{d\mathbb{P}_0} = \exp \left(\int_0^T \mu_s dY_s - \frac{1}{2} \int_0^T \mu_s^2 ds \right).$$

Since the second term is a constant, the Neyman–Pearson test reduces to a threshold on the correlator:

$$\int_0^T \mu_s dY_s \geq \gamma, \quad \gamma \in \mathbb{R}.$$

Remark 20.8. This is the continuous-time matched filter: the optimal detector correlates the observation against the known signal. Nothing about the derivation required guessing the form of the test — Girsanov produced it.

Optimal filtering

Example 20.9. Let $(X_t)_{t \geq 0}$ be an unknown predictable target process, independent of a standard Brownian motion $(B_t)_{t \geq 0}$, observed through

$$dY_t = X_t dt + dB_t.$$

Let $(\mathcal{F}_t^X)$ and $(\mathcal{F}_t^Y)$ be the filtrations generated by X and Y respectively. We want to compute

$$\mathbb{E}_{\mathbb{P}}(X_T \mid \mathcal{F}_T^Y), \quad T \geq 0.$$

Solution. Change measure by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = M_T := \exp \left(- \int_0^T X_s dB_s - \frac{1}{2} \int_0^T X_s^2 ds \right),$$

i.e. apply Girsanov with drift $-X$. Under $\mathbb{Q}$ the observation process Y is itself a standard Brownian motion. Remarkably:

- $\mathcal{F}_T^X$ and $\mathcal{F}_T^Y$ are *independent* under $\mathbb{Q}$;
- $\mathbb{P}|_{\mathcal{F}_T^X} = \mathbb{Q}|_{\mathcal{F}_T^X}$, that is, $\mathbb{P}(A) = \mathbb{Q}(A)$ for all $A \in \mathcal{F}_T^X$ — the law of $(X_t)_{t \in [0, T]}$ is the same under $\mathbb{P}$ and $\mathbb{Q}$.

So under $\mathbb{Q}$ the signal is untouched and the observation carries no information about it, which makes conditional expectations trivial. Transferring back by the change-of-measure (Bayes) formula,

$$\mathbb{E}_{\mathbb{P}}(X_T \mid \mathcal{F}_T^Y) = \frac{\mathbb{E}_{\mathbb{Q}}(M_T^{-1} X_T \mid \mathcal{F}_T^Y)}{\mathbb{E}_{\mathbb{Q}}(M_T^{-1} \mid \mathcal{F}_T^Y)}.$$

Finally, M_T^{-1} is expressible in the observations alone: substituting $\int_0^T X_s dB_s = \int_0^T X_s dY_s - \int_0^T X_s^2 ds$ into the definition of M_T gives

$$M_T^{-1} = \exp \left(\int_0^T X_s dY_s - \frac{1}{2} \int_0^T X_s^2 ds \right).$$

Remark 20.10. The pattern is the same in all three applications: estimation, detection, and pricing each become easy under a measure in which the observation is a plain Brownian motion, and the Radon–Nikodym derivative carries all the structure back.

Lecture 21

Proof of Girsanov's Theorem

Last lecture stated Girsanov's theorem and showed what it buys. This lecture is the proof. We restate it first, since every step below refers back to it.

21.1 Statement

Theorem 21.1 (Girsanov). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry a standard filtration $\mathbb{F}$ and a standard Brownian motion $(B_t)_{t \geq 0}$. Let $\mu \in \mathcal{L}_{\text{loc}}^2$ and*

$$M_t = \exp \left(\underbrace{\int_0^t \mu_s dB_s - \frac{1}{2} \int_0^t \mu_s^2 ds}_{=: Z_t} \right), \quad t \geq 0.$$

If $(M_t)_{t \geq 0}$ is a martingale on $[0, T]$, then $d\mathbb{Q} = M_T d\mathbb{P}$ defines a probability measure $\mathbb{Q} \ll \mathbb{P}$ on $(\Omega, \mathcal{F})$. Moreover

$$X_t = B_t - \int_0^t \mu_s ds$$

is a standard Brownian motion under $\mathbb{Q}$ on the time interval $[0, T]$.

Example 21.2 (What it buys). • Risk-neutral pricing — change the drift to a constant.

- Hypothesis testing — explicit identification of the likelihood ratio.
- Optimal estimation — estimation becomes easy under $\mathbb{Q}$.

Remark 21.3 (Why the diffusion coefficient cannot be changed). **Q.** Why can we not turn $dX_t = \sigma_t dB_t - \mu_t dt$ into a standard Brownian motion when the diffusion σ is nontrivial?

Because quadratic variation is a pathwise, almost sure object, and almost sure statements are preserved under $\mathbb{Q} \ll \mathbb{P}$. Under $\mathbb{P}$ we have $[X]_t = \int_0^t \sigma_s^2 ds$, hence the same holds $\mathbb{Q}$ -a.s.; if this is not identically t then by Lévy's characterization X cannot be a Brownian motion under *any* measure equivalent to $\mathbb{P}$. A change of measure can move drift; it cannot move volatility.

21.2 Proof of Girsanov

Proof. Step 1: $\mathbb{Q}$ is a probability measure. Since $(M_t)_{t \geq 0}$ is assumed to be a martingale on $[0, T]$, $\mathbb{E}(M_T) = \mathbb{E}(M_0) = 1$. Define $\mathbb{Q}(A) := \mathbb{E}_{\mathbb{P}}(M_T \mathbf{1}_A)$ for $A \in \mathcal{F}$. Then

(A1) $\mathbb{Q}(A) \geq 0$, since $M_T > 0$; ✓

(A2) $\mathbb{Q}(\Omega) = \mathbb{E}_{\mathbb{P}}(M_T) = 1$;

(A3) for a disjoint union,

$$\mathbb{Q}\left(\bigsqcup_{i \geq 1} A_i\right) = \mathbb{E}_{\mathbb{P}}(M_T \mathbf{1}_{\bigsqcup A_i}) = \mathbb{E}_{\mathbb{P}}\left(M_T \sum_{i \geq 1} \mathbf{1}_{A_i}\right) \stackrel{\text{MCT}}{=} \sum_{i \geq 1} \mathbb{Q}(A_i).$$

So $\mathbb{Q}$ is a probability measure on $(\Omega, \mathcal{F})$, and $\mathbb{Q} \ll \mathbb{P}$ by construction.

Step 2: XM is a local martingale under $\mathbb{P}$. By the Itô product rule,

$$d(X_t M_t) = X_t dM_t + M_t dX_t + dX_t dM_t.$$

Here $dX_t = -\mu_t dt + dB_t$, and Itô's lemma applied to $M_t = e^{Z_t}$ gives

$$dM_t = M_t dZ_t + \frac{1}{2} M_t \mu_t^2 dt = M_t \mu_t dB_t,$$

since $dZ_t = \mu_t dB_t - \frac{1}{2} \mu_t^2 dt$ and $d[Z]_t = \mu_t^2 dt$. Hence $dX_t dM_t = \mu_t M_t dt$ and

$$\begin{aligned} d(X_t M_t) &= X_t M_t \mu_t dB_t + M_t (-\mu_t dt + dB_t) + \mu_t M_t dt \\ &= (X_t M_t \mu_t + M_t) dB_t = (1 + X_t \mu_t) M_t dB_t, \end{aligned}$$

so $X_t M_t$ is a local martingale under $\mathbb{P}$.

Step 3: X is a local martingale under $\mathbb{Q}$. Let

$$\tau_k := \inf\{t \geq 0 : |X_t M_t| \geq k\},$$

so that $(X_{t \wedge \tau_k} M_{t \wedge \tau_k})_{t \geq 0}$ is a bounded, hence honest, martingale under $\mathbb{P}$. Fix $k \geq 1$, $0 \leq s \leq t \leq T$, and $A \in \mathcal{F}_{s \wedge \tau_k} \subseteq \mathcal{F}_{t \wedge \tau_k}$. Then

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}(X_{t \wedge \tau_k} \mathbf{1}_A) &= \mathbb{E}_{\mathbb{P}}(M_T X_{t \wedge \tau_k} \mathbf{1}_A) \\ &= \mathbb{E}_{\mathbb{P}}\left(\mathbb{E}_{\mathbb{P}}(M_T \mid \mathcal{F}_{t \wedge \tau_k}) X_{t \wedge \tau_k} \mathbf{1}_A\right) \\ &= \mathbb{E}_{\mathbb{P}}(M_{t \wedge \tau_k} X_{t \wedge \tau_k} \mathbf{1}_A) && \text{(optional stopping for } M) \\ &= \mathbb{E}_{\mathbb{P}}\left(\mathbb{E}_{\mathbb{P}}(M_{t \wedge \tau_k} X_{t \wedge \tau_k} \mid \mathcal{F}_{s \wedge \tau_k}) \mathbf{1}_A\right) \\ &= \mathbb{E}_{\mathbb{P}}(M_{s \wedge \tau_k} X_{s \wedge \tau_k} \mathbf{1}_A) && \text{(Step 2, stopped at } \tau_k) \\ &= \mathbb{E}_{\mathbb{P}}\left(\mathbb{E}_{\mathbb{P}}(M_T \mid \mathcal{F}_{s \wedge \tau_k}) X_{s \wedge \tau_k} \mathbf{1}_A\right) \\ &= \mathbb{E}_{\mathbb{P}}(M_T X_{s \wedge \tau_k} \mathbf{1}_A) = \mathbb{E}_{\mathbb{Q}}(X_{s \wedge \tau_k} \mathbf{1}_A). \end{aligned}$$

That is,

$$\mathbb{E}_{\mathbb{Q}}(X_{t \wedge \tau_k} \mid \mathcal{F}_{s \wedge \tau_k}) = X_{s \wedge \tau_k}.$$

We want the same statement with $\mathcal{F}_s$ in place of $\mathcal{F}_{s \wedge \tau_k}$. Split on whether τ_k has already occurred:

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}(X_{t \wedge \tau_k} \mid \mathcal{F}_s) &= \mathbb{E}_{\mathbb{Q}}(X_{t \wedge \tau_k} \mathbf{1}_{\{\tau_k \leq s\}} \mid \mathcal{F}_s) + \mathbb{E}_{\mathbb{Q}}(X_{t \wedge \tau_k} \mathbf{1}_{\{\tau_k > s\}} \mid \mathcal{F}_s) \\ &= X_{s \wedge \tau_k} \mathbf{1}_{\{\tau_k \leq s\}} + \mathbb{E}_{\mathbb{Q}}(X_{t \wedge \tau_k} \mid \mathcal{F}_s) \mathbf{1}_{\{\tau_k > s\}} \\ &= X_{s \wedge \tau_k} \mathbf{1}_{\{\tau_k \leq s\}} + \mathbb{E}_{\mathbb{Q}}(X_{t \wedge \tau_k} \mid \mathcal{F}_{s \wedge \tau_k}) \mathbf{1}_{\{\tau_k > s\}} \\ &= X_{s \wedge \tau_k}. \end{aligned}$$

On $\{\tau_k \leq s\}$ we used $X_{t \wedge \tau_k} = X_{\tau_k} = X_{s \wedge \tau_k}$, which is $\mathcal{F}_s$ -measurable there; on $\{\tau_k > s\} \in \mathcal{F}_{s \wedge \tau_k}$ the two conditionings agree. So $(X_t)_{t \geq 0}$ is a continuous local martingale under $\mathbb{Q}$, with localizing sequence $(\tau_k)_{k \geq 1}$.

Step 4: identify the quadratic variation and apply Lévy. Under $\mathbb{P}$, X has diffusion coefficient 1, so

$$[X]_t = \int_0^t 1 \, ds = t.$$

Quadratic variation is defined as a limit in probability, and $\mathbb{Q} \ll \mathbb{P}$ preserves such limits, so $[X]_t = t$ also under $\mathbb{Q}$. Finally $(\Omega, \mathcal{F}, \mathbb{Q})$ is a complete probability space carrying the standard filtration $\mathbb{F}$, and X is a continuous local martingale under $\mathbb{Q}$ with $X_0 = 0$. By Lévy's characterization, X is a standard Brownian motion under $\mathbb{Q}$. $\square$

Remark 21.4 (The hypothesis we have not checked). Everything above runs on the standing assumption that $(M_t)_{t \geq 0}$ is a *martingale* on $[0, T]$ — equivalently, that $\mathbb{E}(M_T) = 1$. That is a genuine hypothesis, not a formality: M is always a nonnegative continuous local martingale, hence a supermartingale, and $\mathbb{E}(M_T) < 1$ is possible. Without it there is no $\mathbb{Q}$ to speak of, since $\mathbb{Q}(\Omega) = \mathbb{E}(M_T) \neq 1$. Next lecture we give the standard checkable sufficient condition.

Lecture 22

Novikov's Condition and SDEs

22.1 The missing hypothesis

To apply Girsanov's theorem we must first establish the martingale property of

$$M_t = \exp \left(\int_0^t \mu_s dB_s - \frac{1}{2} \int_0^t \mu_s^2 ds \right), \quad t \geq 0.$$

This is not automatic. Itô's lemma applied to $f(x) = e^x$ gives $dM_t = \mu_t M_t dB_t$, so M is always a nonnegative continuous local martingale and therefore a supermartingale; but a nonnegative supermartingale can lose mass, and $\mathbb{E}(M_T) < 1$ is possible. Since $\mathbb{Q}(\Omega) = \mathbb{E}(M_T)$, without this hypothesis there is no probability measure $\mathbb{Q}$ to change to.

The most common sufficient condition is the following.

Theorem 22.1 (Novikov). *Fix $T > 0$ and let $\mu \in \mathcal{L}_{\text{loc}}^2$ satisfy*

$$\mathbb{E} \left(\exp \left(\frac{1}{2} \int_0^T \mu_s^2 ds \right) \right) < \infty.$$

Then $(M_t)_{t \geq 0}$ is a martingale on $[0, T]$.

22.2 Proof of Novikov's condition

Proof ("lazy man's Novikov"). We prove the theorem under the slightly stronger hypothesis that

$$\mathbb{E} \left(\exp \left(\frac{1+\varepsilon}{2} \int_0^T \mu_s^2 ds \right) \right) < \infty \quad \text{for some } \varepsilon > 0. \quad (22.1)$$

The sharp constant $\frac{1}{2}$ is harder and is not what we are after here; the point of the argument is the mechanism, which is a uniform L^α bound on the stopped exponential.

Step 1: a family of exponential local martingales. For $\beta \in \mathbb{R}$ write

$$M_t(\beta) := \exp \left(\beta \int_0^t \mu_s dB_s - \frac{\beta^2}{2} \int_0^t \mu_s^2 ds \right),$$

so that $M = M(1)$. Applying Itô's lemma to $f(x) = e^x$ as above gives $dM_t(\beta) = \beta \mu_t M_t(\beta) dB_t$, so each $M(\beta)$ is a nonnegative continuous local martingale, hence a supermartingale. In particular, for *every* stopping time τ ,

$$\mathbb{E}(M_{T \wedge \tau}(\beta)) \leq \mathbb{E}(M_0(\beta)) = 1. \quad (22.2)$$

Step 2: splitting the α -th power. Fix $\alpha > 1$ and conjugate exponents $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Note $\alpha p > 1$. Split $M_{T \wedge \tau}^\alpha$ so that one factor is again a member of the family $M(\cdot)$:

$$M_{T \wedge \tau}^\alpha = \underbrace{\exp \left(\alpha \int_0^{T \wedge \tau} \mu_s dB_s - \frac{p\alpha^2}{2} \int_0^{T \wedge \tau} \mu_s^2 ds \right)}_U \cdot \underbrace{\exp \left(\frac{\alpha(\alpha p - 1)}{2} \int_0^{T \wedge \tau} \mu_s^2 ds \right)}_V,$$

an identity because $-\frac{p\alpha^2}{2} + \frac{\alpha(\alpha p - 1)}{2} = -\frac{\alpha}{2}$. The point of the splitting is that $U^p = M_{T \wedge \tau}(p\alpha)$ exactly, so Hölder's inequality with exponents p, q gives

$$\begin{aligned} \mathbb{E}(M_{T \wedge \tau}^\alpha) &\leq \mathbb{E}(U^p)^{1/p} \mathbb{E}(V^q)^{1/q} \\ &= \mathbb{E}(M_{T \wedge \tau}(p\alpha))^{1/p} \mathbb{E} \left(\exp \left(\frac{q\alpha(\alpha p - 1)}{2} \int_0^{T \wedge \tau} \mu_s^2 ds \right) \right)^{1/q} \\ &\leq \mathbb{E} \left(\exp \left(\frac{\alpha p(\alpha p - 1)}{2(p - 1)} \int_0^T \mu_s^2 ds \right) \right)^{1/q}, \end{aligned}$$

using (22.2) on the first factor, then $q = \frac{p}{p-1}$ on the second. Enlarging $T \wedge \tau$ to T in the last step is legitimate because the exponent is nonnegative, $\alpha p > 1$.

Step 3: choosing the exponents. Take $p = 1 + \frac{\varepsilon}{2}$. As $\alpha \downarrow 1$ the coefficient

$$\frac{\alpha p(\alpha p - 1)}{p - 1} \longrightarrow \frac{p(p - 1)}{p - 1} = p = 1 + \frac{\varepsilon}{2} < 1 + \varepsilon,$$

so we may fix $\alpha > 1$ with $\frac{\alpha p(\alpha p - 1)}{p - 1} \leq 1 + \varepsilon$. With that choice, (22.1) gives

$$\mathbb{E}(M_{T \wedge \tau}^\alpha) \leq \mathbb{E} \left(\exp \left(\frac{1 + \varepsilon}{2} \int_0^T \mu_s^2 ds \right) \right)^{1/q} =: C < \infty.$$

So there exists $\alpha > 1$ with $\mathbb{E}(M_{T \wedge \tau}^\alpha) \leq C < \infty$ for *every* stopping time τ — and, crucially, C does not depend on τ .

Step 4: uniform integrability closes the gap. Let $(\tau_k)_{k \geq 1}$ be a localizing sequence for $(M_t)_{t \geq 0}$, so that each $M_{T \wedge \tau_k}$ has expectation 1. By Step 3 the family $(M_{T \wedge \tau_k})_{k \geq 1}$ is bounded in L^α for some $\alpha > 1$, hence uniformly integrable. It also converges almost surely, $M_{T \wedge \tau_k} \rightarrow M_T$ as $k \rightarrow \infty$, by continuity of M and $\tau_k \rightarrow \infty$. Uniform integrability upgrades almost sure convergence to convergence in L^1 , so

$$\mathbb{E}(M_T) = \lim_{k \rightarrow \infty} \mathbb{E}(M_{T \wedge \tau_k}) = 1.$$

A supermartingale whose terminal expectation equals its initial value is a martingale, so $(M_t)_{t \geq 0}$ is a martingale on $[0, T]$. $\square$

Remark 22.2. The shape of this argument is worth remembering on its own: a nonnegative local martingale is always a supermartingale, so the only thing that can go wrong is a loss of mass in the limit, and *any* uniform integrability estimate on the localized process rules that out. Novikov's condition is simply the most convenient hypothesis that produces one.

22.3 The change of measure does not extend to an infinite horizon

Every statement so far has been over a *finite* interval $[0, T]$. That restriction is not an artifact of the proof.

Example 22.3. Let

$$dX_t = -\theta dt + dB_t, \quad \theta \in \mathbb{R},$$

so $X_t = B_t - \theta t$, and suppose $\theta \neq 0$. By the strong law for Brownian motion,

$$\frac{X_t}{t} = \frac{B_t}{t} - \theta \xrightarrow{\text{a.s.}} -\theta \neq 0 \quad \text{as } t \rightarrow \infty,$$

so, writing $A := \{\lim_{t \rightarrow \infty} X_t/t = 0\}$, we have $\mathbb{P}(A) = 0$. If $\mathbb{Q} \ll \mathbb{P}$, then A is $\mathbb{Q}$ -null as well:

$$\mathbb{Q}\left(\lim_{t \rightarrow \infty} \frac{X_t}{t} = 0\right) = 0.$$

But a standard Brownian motion W satisfies $W_t/t \rightarrow 0$ almost surely, so X being an SBM under $\mathbb{Q}$ would force $\mathbb{Q}(A) = 1$. Hence $(X_t)_{t \geq 0}$ cannot be a standard Brownian motion under any $\mathbb{Q} \ll \mathbb{P}$, and the change of measure cannot be extended over an infinite time horizon.

Remark 22.4. On $[0, T]$ the two measures are mutually absolutely continuous and see exactly the same null events. Over the infinite horizon the drift becomes detectable from a single path with probability one — the long-run slope X_t/t estimates $-\theta$ exactly — and the two measures become mutually singular. No reweighting can hide a constant drift forever.

22.4 Stochastic differential equations

With the Itô integral, Itô's lemma, and the change-of-measure machinery in place, we can return to the equations that motivated the course. Consider

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t, \quad X_0 \text{ given}, \quad t \in [0, T]. \quad (\text{SDE})$$

Q. When does there exist a process X solving this SDE? Is the solution unique?

Note the change in character. Until now μ and σ were *given* processes and X was *defined* by integrating them, so there was nothing to solve. In (SDE) the coefficients are functions of the current state, so the equation is genuinely implicit: X appears on both sides. Answering both questions is the subject of the next lecture, and the answer is the exact stochastic analogue of the Picard–Lindelöf theorem for ODEs.

Lecture 23

Existence and Uniqueness for SDEs

23.1 What a solution is

Given $\mu, \sigma: (t, x) \mapsto \mu(t, x), \sigma(t, x)$, we are interested in whether there is a process X solving

$$X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s, \quad t \in [0, T], \quad \text{a.s.} \quad (\star)$$

At a minimum X should be continuous and adapted — continuity because the integrals on the right are, and adaptedness because otherwise the stochastic integral is not even defined.

Theorem 23.1. *Suppose $X_0 \in L^2$ is $\mathcal{F}_0$ -measurable and there is $k > 0$ such that, for all $t \in [0, T]$ and $x, y \in \mathbb{R}$,*

$$|\mu(t, x) - \mu(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq k|x - y|, \quad (23.1)$$

$$|\mu(t, x)| + |\sigma(t, x)| \leq k(1 + |x|). \quad (23.2)$$

Then there is a continuous, L^2 -bounded solution $(X_t)_{t \in [0, T]}$ of $(\star)$, adapted to the filtration generated by X_0 and $(B_t)_{t \geq 0}$. Moreover any other continuous adapted solution X' of $(\star)$ is indistinguishable from X :

$$\mathbb{P}(X_t = X'_t \forall t \in [0, T]) = 1.$$

Conditions (23.1) and (23.2) are the exact analogues of the Picard–Lindelöf hypotheses for ODEs: Lipschitz continuity buys uniqueness, linear growth prevents blow-up in finite time.

Lemma 23.2 (Grönwall's inequality). *(a) If u is differentiable on (a, b) and $u'(t) \leq \beta(t)u(t)$ there, then*

$$u(t) \leq u(a) \exp \left(\int_a^t \beta(s) ds \right).$$

(b) If $u(t) \leq \alpha(t) + \int_a^t \beta(s)u(s) ds$ with α nondecreasing and $\beta \geq 0$, then

$$u(t) \leq \alpha(t) \exp \left(\int_a^t \beta(s) ds \right).$$

23.2 An a priori L^2 bound

Proof of Theorem 23.1: the a priori bound. Suppose $(X_t)_{t \in [0, T]}$ is a continuous adapted solution of $(\star)$. Then $(s, \omega) \mapsto \sigma(s, X_s(\omega)) \in \mathcal{L}_{\text{loc}}^2$ and $(s, \omega) \mapsto \mu(s, X_s(\omega)) \in \mathcal{L}_{\text{loc}}^1$, since both are continuous in s along each path. Localize by

$$\tau_n := \inf\{t \geq 0 : |X_t| \geq n\}.$$

Then, using $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$,

$$\begin{aligned} \mathbb{E}(X_{t \wedge \tau_n}^2) &= \mathbb{E} \left(\left(X_0 + \int_0^{t \wedge \tau_n} \mu(s, X_{s \wedge \tau_n}) ds + \int_0^{t \wedge \tau_n} \sigma(s, X_{s \wedge \tau_n}) dB_s \right)^2 \right) \\ &\lesssim \mathbb{E}(X_0^2) + \mathbb{E} \left(\left(\int_0^{t \wedge \tau_n} \mu(s, X_{s \wedge \tau_n}) ds \right)^2 \right) + \mathbb{E} \left(\left(\int_0^{t \wedge \tau_n} \sigma(s, X_{s \wedge \tau_n}) dB_s \right)^2 \right) \\ &\leq \mathbb{E}(X_0^2) + T \int_0^t \mathbb{E}(\mu(s, X_{s \wedge \tau_n})^2) ds + \mathbb{E} \left(\int_0^t \sigma(s, X_{s \wedge \tau_n})^2 ds \right), \end{aligned}$$

the middle term by Cauchy–Schwarz and the last by the Itô isometry. Applying the linear growth bound (23.2) to both integrands,

$$\mathbb{E}(X_{t \wedge \tau_n}^2) \leq C + C \int_0^t \mathbb{E}(X_{s \wedge \tau_n}^2) ds$$

for a constant $C = C(T, k, \mathbb{E}(X_0^2))$, notably *independent of n* . The map $u: t \mapsto \mathbb{E}(X_{t \wedge \tau_n}^2)$ is continuous by dominated convergence, so Grönwall gives

$$\mathbb{E}(X_{t \wedge \tau_n}^2) \leq C e^{CT} \quad \forall n.$$

Since $\tau_n \rightarrow \infty$ a.s. and X is continuous, $X_{t \wedge \tau_n} \rightarrow X_t$ a.s., so by Fatou

$$\mathbb{E}(X_t^2) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(X_{t \wedge \tau_n}^2) \leq C e^{CT}.$$

So every continuous adapted solution is automatically L^2 -bounded on $[0, T]$. $\square$

23.3 Uniqueness

Proof of Theorem 23.1: uniqueness. Let $(Y_t)_{t \geq 0}$ be any other continuous adapted solution of $(\star)$ with $Y_0 = X_0$. By the previous step both X and Y are L^2 -bounded, so all the expectations below are finite. Subtracting the two integral equations,

$$\begin{aligned} \mathbb{E}((X_t - Y_t)^2) &\lesssim T \int_0^t \mathbb{E}((\mu(s, X_s) - \mu(s, Y_s))^2) ds + \mathbb{E} \left(\int_0^t (\sigma(s, X_s) - \sigma(s, Y_s))^2 ds \right) \\ &\leq K \int_0^t \mathbb{E}((X_s - Y_s)^2) ds, \end{aligned}$$

the last line by the Lipschitz condition (23.1), with $K = K(T, k)$. This is Grönwall's inequality with $\alpha \equiv 0$, so

$$\mathbb{E}((X_t - Y_t)^2) \leq 0 \cdot e^{KT} = 0 \implies X_t = Y_t \text{ a.s.}$$

for each fixed $t \in [0, T]$. Taking the union of the exceptional sets over rational t and using that both paths are continuous upgrades this to $\mathbb{P}(X_t = Y_t \forall t \in [0, T]) = 1$, i.e. X and Y are indistinguishable. $\square$

23.4 Existence: Picard iteration

Start from the constant process $X_t^{(0)} = X_0$, $t \in [0, T]$, and define recursively

$$X_t^{(n+1)} = X_0 + \int_0^t \mu(s, X_s^{(n)}) ds + \int_0^t \sigma(s, X_s^{(n)}) dB_s, \quad t \in [0, T].$$

First, every iterate is square integrable. By linear growth,

$$\mathbb{E}\left((X_t^{(1)})^2\right) \leq A + B \int_0^t \mathbb{E}(X_0^2) ds \leq A + BT \mathbb{E}(X_0^2),$$

and inductively $\mathbb{E}((X_t^{(n)})^2) \leq A_n + B_n \mathbb{E}(X_0^2)$ for constants A_n, B_n depending on n, T and k .

Now write $\Delta_t^{(n)} := X_t^{(n+1)} - X_t^{(n)}$. Subtracting consecutive iterates and using the Lipschitz condition exactly as in the uniqueness argument,

$$\mathbb{E}\left(|\Delta_t^{(n)}|^2\right) \leq C \int_0^t \mathbb{E}\left(|\Delta_s^{(n-1)}|^2\right) ds \leq C^2 \int_0^t \int_0^s \mathbb{E}\left(|\Delta_u^{(n-2)}|^2\right) du ds \lesssim \frac{(CT)^n}{n!},$$

iterating n times: each pass contributes a factor C and shrinks the simplex of integration, and $|\{0 \leq u_1 \leq \dots \leq u_n \leq T\}| = T^n/n!$.

Upgrading to a bound on the supremum

For the a.s. convergence argument we will need the stronger estimate

$$\mathbb{E}\left(\sup_{t \in [0, T]} |\Delta_t^{(n)}|^2\right) \lesssim \frac{(CT)^n}{n!}.$$

Write $\delta_s^{(n)} := \sigma(s, X_s^{(n)}) - \sigma(s, X_s^{(n-1)})$ for the difference of the diffusion coefficients along consecutive iterates. Splitting the drift and diffusion parts,

$$\begin{aligned} \sup_{t \in [0, T]} |\Delta_t^{(n)}|^2 &\lesssim T \int_0^T \left| \mu(s, X_s^{(n)}) - \mu(s, X_s^{(n-1)}) \right|^2 ds \\ &\quad + \sup_{t \in [0, T]} \left| \int_0^t \delta_s^{(n)} dB_s \right|^2. \end{aligned}$$

The first term is already deterministic in t . For the second, the integral is a continuous L^2 -martingale in t , so Doob's L^2 maximal inequality followed by the Itô isometry gives

$$\mathbb{E}\left(\sup_{t \in [0, T]} \left| \int_0^t \delta_s^{(n)} dB_s \right|^2\right) \leq 4 \mathbb{E}\left(\int_0^T (\delta_s^{(n)})^2 ds\right).$$

Applying (23.1) to both terms,

$$\mathbb{E}\left(\sup_{t \in [0, T]} |\Delta_t^{(n)}|^2\right) \lesssim \mathbb{E}\left(\int_0^T |\Delta_s^{(n-1)}|^2 ds\right) \lesssim \frac{(CT)^n}{n!}.$$

We finish the existence argument next lecture: the factorial decay is summable, which via Borel–Cantelli gives uniform convergence of the iterates along every path.

Lecture 24

Picard Iteration and Euler–Maruyama

24.1 Recap

Theorem 24.1. *Let $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dB_t$ with X_0 $\mathcal{F}_0$ -measurable. If μ and σ satisfy the linear growth and Lipschitz conditions on $t \in [0, T]$, and $X_0 \in L^2$, then there is a unique continuous adapted process solving the SDE, and it is L^2 -bounded:*

$$X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s \quad \forall t \in [0, T], \text{ a.s.} \quad (24.1)$$

24.2 Existence, completed

Recall the Picard iteration: $X_t^{(0)} = X_0$ and

$$X_t^{(n+1)} = X_0 + \int_0^t \mu(s, X_s^{(n)}) ds + \int_0^t \sigma(s, X_s^{(n)}) dB_s, \quad t \in [0, T].$$

Each iterate is continuous and adapted to the filtration generated by X_0 and $(B_t)_{t \geq 0}$, each is L^2 -bounded, and from last lecture

$$\mathbb{E} \left(\sup_{t \in [0, T]} |X_t^{(n+1)} - X_t^{(n)}|^2 \right) \lesssim \frac{C^n}{n!}.$$

Almost sure uniform convergence

By Chebyshev,

$$\mathbb{P} \left(\sup_{t \in [0, T]} |X_t^{(n+1)} - X_t^{(n)}| \geq 2^{-n} \right) \leq 4^n \mathbb{E} \left(\sup_{t \in [0, T]} |X_t^{(n+1)} - X_t^{(n)}|^2 \right) \lesssim \frac{(4C)^n}{n!}.$$

These probabilities are summable, since $\sum_n (4C)^n / n! = e^{4C} < \infty$, so Borel–Cantelli gives

$$\mathbb{P} \left\{ \sup_{t \in [0, T]} |X_t^{(n+1)} - X_t^{(n)}| \geq 2^{-n} \text{ i.o.} \right\} = 0.$$

Almost surely, then, the tail of the telescoping series is dominated by $\sum_n 2^{-n}$, hence converges uniformly on $[0, T]$. Therefore

$$X_t := \lim_{n \rightarrow \infty} X_t^{(n)} = X_t^{(0)} + \sum_{n \geq 0} (X_t^{(n+1)} - X_t^{(n)})$$

exists, and X is continuous a.s. as a uniform limit of continuous processes.

Passing to the limit in the equation

The same estimate controls the L^2 distance to the limit:

$$\|X_t - X_t^{(n)}\|_2 \leq \sum_{k \geq n} \|X_t^{(k+1)} - X_t^{(k)}\|_2 \lesssim \sum_{k \geq n} \sqrt{\frac{C^k}{k!}} \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

the series converging because $\sqrt{C^k/k!}$ decays faster than geometrically. Hence, by the Lipschitz condition together with Cauchy–Schwarz and the Itô isometry,

$$\int_0^t \mu(s, X_s^{(n)}) ds + \int_0^t \sigma(s, X_s^{(n)}) dB_s \xrightarrow{L^2} \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s.$$

But the left-hand side is exactly $X_t^{(n+1)} - X_0$, which converges a.s. to $X_t - X_0$. Identifying the two limits,

$$X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s \quad \forall t \in [0, T],$$

which is (24.1). □

24.3 The Euler–Maruyama scheme

Existence and uniqueness say the solution is there; they do not say how to compute it. Consider the autonomous SDE

$$dX_t = \mu(X_t) dt + \sigma(X_t) dB_t, \quad X_0 = x_0,$$

with μ, σ Lipschitz, so that Theorem 24.1 gives a unique continuous adapted solution $(X_t)_{t \geq 0}$. Fix a horizon $[0, T]$ and discretize:

$$t_n = \frac{nT}{N}, \quad \Delta t = \frac{T}{N}.$$

Definition 24.2 (Euler–Maruyama). Initialize $Y_0^{(N)} = x_0$, and for $n = 1, \dots, N$ set

$$Y_{t_n}^{(N)} = Y_{t_{n-1}}^{(N)} + \mu(Y_{t_{n-1}}^{(N)}) \Delta t + \sigma(Y_{t_{n-1}}^{(N)}) \underbrace{(B_{t_n} - B_{t_{n-1}})}_{\substack{i.i.d. \\ \sim \mathcal{N}(0, \Delta t)}}.$$

This is the obvious discretization — freeze the coefficients at the left endpoint of each subinterval, exactly as in the definition of the Itô integral, and simulate the Brownian increments directly.

Theorem 24.3 (Strong convergence). *If μ, σ are k -Lipschitz, there exists $C = C(T, k, x_0)$ such that*

$$\mathbb{E}\left(|X_{t_n} - Y_{t_n}^{(N)}|^2\right) \leq C\Delta t \quad \forall n \in \{0, \dots, N\}.$$

Proof. Extend the scheme to a piecewise constant process,

$$\bar{Y}_t := Y_{t_n}^{(N)} \quad \text{for } t \in [t_n, t_{n+1}),$$

and define the associated continuous process

$$Y_t := x_0 + \int_0^t \mu(\bar{Y}_s) ds + \int_0^t \sigma(\bar{Y}_s) dB_s.$$

By construction $Y_{t_n} = \bar{Y}_{t_n} = Y_{t_n}^{(N)}$, so it suffices to bound $\mathbb{E}(|X_t - Y_t|^2)$.

Subtracting the two integral equations and using the Lipschitz condition together with Cauchy–Schwarz and the Itô isometry,

$$\begin{aligned} \mathbb{E}(|X_t - Y_t|^2) &\lesssim \mathbb{E}\left(\int_0^t (\mu(X_s) - \mu(\bar{Y}_s))^2 ds\right) + \mathbb{E}\left(\left|\int_0^t (\sigma(X_s) - \sigma(\bar{Y}_s)) dB_s\right|^2\right) \\ &\lesssim \int_0^t \mathbb{E}\left((X_s - \bar{Y}_s)^2\right) ds \\ &\lesssim \int_0^t \mathbb{E}\left((X_s - Y_s)^2\right) ds + \int_0^t \mathbb{E}\left((Y_s - \bar{Y}_s)^2\right) ds. \end{aligned}$$

For the last term, fix $s \in [t_n, t_{n+1})$. Since $\bar{Y}$ is constant on that subinterval,

$$Y_s - \bar{Y}_s = (s - t_n)\mu(Y_{t_n}) + \sigma(Y_{t_n})(B_s - B_{t_n}),$$

so

$$\mathbb{E}\left(|Y_s - \bar{Y}_s|^2\right) \lesssim \mathbb{E}\left((s - t_n)^2 \mu(Y_{t_n})^2\right) + 2(s - t_n) \mathbb{E}\left(|\sigma(Y_{t_n})|^2\right) \lesssim (\Delta t)^2 + \Delta t \lesssim \Delta t,$$

using that the scheme is L^2 -bounded uniformly in N — a consequence of the linear growth implied by Lipschitz continuity. Substituting back,

$$\mathbb{E}(|X_t - Y_t|^2) \lesssim \Delta t + \int_0^t \mathbb{E}\left((X_s - Y_s)^2\right) ds,$$

and Grönwall’s inequality gives $\mathbb{E}(|X_t - Y_t|^2) \leq C\Delta t$ with $C = C(T, k, x_0)$. Evaluating at $t = t_n$, where $Y_{t_n} = Y_{t_n}^{(N)}$, gives the claim. $\square$

Remark 24.4. The scheme has *strong order* $\frac{1}{2}$: the root-mean-square error scales like $\sqrt{\Delta t}$, not Δt . That is the price of the Brownian increments, whose own size is $\Theta(\sqrt{\Delta t})$ — the same $\sqrt{dt}$ scaling that forced the second-order term in Itô’s lemma. Higher-order schemes exist, but they require simulating iterated stochastic integrals rather than plain increments.

Remark 24.5 (Where this leaves us). We now have all three pieces promised in Lecture 1: the equation $dX_t = u_t(X_t) dt + dB_t$ has a precise meaning (the Itô integral), a unique solution (Theorem 24.1), and a law we can access — analytically through Fokker–Planck, and numerically through Euler–Maruyama.